Measuring the Gap: Semantic Alignment Between AI-for-Sustainability Research and SDG Policy Frameworks

Contents

Abstract

Shared SDG labels do not establish that research and policy *frame* a goal similarly (Armitage et al., 2020). This paper contributes a measurement framework that separates research-policy divergence into two independent dimensions — coverage (which SDGs each corpus prioritises) and within-SDG semantic framing (how differently the same SDG is discussed) — and isolates topic from register using Iterative Nullspace Projection (INLP). Using frozen Sentence-BERT embeddings and a supervised logistic-regression classifier trained on 52,835 SDG-labelled reference texts (test macro-F_1 = 0.816), I score 3,105,144 segments from 2,536,771 AI-for-sustainability abstracts and 40,597 segments from 6,367 institutional policy sources, and measure both dimensions across all 17 SDGs.

The raw coverage–semantic-gap correlation is null ($\rho = -0.012$, $p = 0.963$), but decomposing within-SDG distance into topic and register components reveals a systematic cancellation: the register-adjusted topic gap is positively associated with coverage ($\rho = 0.544$, $p = 0.024$) while the removed register component is negatively associated ($\rho = -0.390$, $p = 0.122$). At $n = 17$ SDGs only the adjusted association reaches significance; the register component is directionally negative but not significant, so the null raw gap reflects a partial, not complete, cancellation.

The result is nonetheless stable: the adjusted semantic-gap ranking survives every sensitivity check — encoders, classifier family, retrieval strategy, segment cap, and policy-source family — and distribution-aware metrics confirm it is a mean-direction effect. Coverage and framing should be measured and reported as separate axes of divergence; shared labels are insufficient evidence of shared framing (Heilinger et al., 2024).

Keywords: research-policy alignment; Sustainable Development Goals; semantic similarity; transformer embeddings; measurement protocol; bibliometrics.

Data and code availability. The analysis code and the frozen, processed outputs are version-controlled at https://github.com/qmanhbeo/dissertation-bham. The repository is private during thesis examination and will be made public after grade release; raw upstream data are retrieved from the OpenAlex API under its terms of use.

Competing interests. The author declares none.

1 Introduction

Research-policy alignment is often inferred from whether scientific and policy texts address the same topics (Bornmann et al., 2016), yet shared topical coverage does not establish comparable framing: a paper and a policy document can both invoke SDG 13 (Climate Action) or SDG 3 (Good Health) while emphasising different problems, actors, and interventions. High-level thematic agreement can therefore mask deeper divergence in how each discourse community approaches the same goal.

Existing methods for mapping research to the Sustainable Development Goals (SDGs) operate mainly at the level of lexical or bibliometric correspondence: keyword dictionaries, Boolean queries, or citation and mention counts (Armitage et al., 2020; Kashnitsky et al., 2024). They measure whether a text is *about* a given SDG, not how it frames it. That interchangeability is what this paper questions.

This paper proposes that research-policy alignment has at least two separable dimensions: coverage divergence (which SDGs each corpus prioritises) and semantic framing divergence (how differently they discuss the same SDG). Its contribution is a measurement framework. I advance AI–SDG mapping from Level 1 (lexical correspondence) to Level 2 (observed textual-semantic proximity in embedding space; Section 2.2), treating semantic proximity as a register-sensitive proxy for framing, not evidence of substantive alignment. I operationalise both dimensions across all 17 SDGs with frozen Sentence-BERT embeddings (Reimers & Gurevych, 2019) and a supervised logistic-regression classifier, and stress-test the instrument against its principal measurement choices, reporting every gap ranking under a cross-sensitivity grid of assignment method, policy source family, and segment cap, with sample-stability and register-adjustment checks.

The near-zero raw coverage-vs-semantic-gap correlation may reflect cancellation: topic divergence rises with coverage divergence while register divergence falls with it, and the two appear to offset ($p = 0.024$ and $p = 0.122$ respectively, $n = 17$ SDGs). This matters because SDG tagging is now institutionalised: OpenAlex labels works, funders count SDG outputs, and journals print SDG badges (Heilinger et al., 2024; Heras-Saizarbitoria et al., 2022); if shared labels mask a register–topic cancellation, that threatens how the SDG-monitoring ecosystem reads its own signals. After register removal, the adjusted topic gap identifies SDG 17 as the most divergent goal and the one most sensitive to how divergence is measured; under the raw gap (naive baseline), SDG 13 leads instead. SDG 15 is the stably least-divergent goal under both. I report this pattern with its scope conditions.

The central research question is: *To what extent do academic AI-for-sustainability research and international institutional SDG policy discourse differ in SDG coverage and semantic framing across the 17 SDGs, and how are these two dimensions related?* Section 2 reviews the literature and introduces the three-level framework, Section 3 the corpora and measurement procedures, and Sections 4 and 5 the findings and their interpretation.

2 Literature Review

2.1 AI and the SDGs: What Bibliometrics Shows

Existing assessments of AI and the SDGs rest on expert views or publication counts rather than on the texts themselves (Vinuesa et al., 2020). Broader SDG scientometrics and the AI-specific inventories converge on one robust finding: SDG 3 (Good Health) and SDG 7 (Affordable and Clean Energy) are the most addressed goals in the AI-specific studies, followed by SDG 4, 13, 11, and 16. The evidence base spans a citation-based corpus of roughly 25,000 publications on the Millennium and Sustainable Development Goals (M&SDGs) (Bautista-Puig et al., 2021), 20,511 AI publications (2001–2020) (Singh et al., 2024), 108 AI-for-social-good projects (Cowls et al., 2021), and 1,288 big-data and AI publications (Nedungadi et al., 2024), together with a recent review of 792 AI applications in SDG-related research (Gohr et al., 2025). The least-covered goals differ between the AI-specific inventories and the broader bibliometric picture, and none of these studies measures within-SDG semantic framing (Gohr et al., 2025).

SDG publication mapping is methodologically contingent. Armitage et al. (2020) shows that what counts as SDG-relevant depends on interpretive choices about queries, labels, and databases; Kashnitsky et al. (2024) reach the same caution from a broader comparison of Boolean, machine-learning, and hybrid mapping systems, finding no single best approach and no single "golden" validation set. The dependence persists in AI-based mapping: keyword baselines still omit publications that never state an SDG keyword explicitly, so the reported corpus boundary remains query-dependent (Yin et al., 2025). The embedding-based approach used here reduces some vocabulary-level brittleness of keyword dictionaries, but its results remain conditional on corpus construction, centroid provenance, and model choice.

The SDGs themselves are interdependent by design, yet no operational framework specifies how (Nilsson et al., 2016); single-SDG assignment is therefore a simplifying assumption. One goal breaks the pattern. Le Blanc (2015), in a network analysis of the proposed SDGs, treats SDG 17 (Partnerships for the Goals) as the dedicated goal for cross-cutting means of implementation: finance, trade, technology transfer, and capacity building, excluding it from cross-goal mapping because its targets are implementation infrastructure rather than substantive domains. The Griggs et al. (2017) guide likewise describes SDG 17 as the main enabler of the whole framework. Unlike every other SDG, whose discourse centres on a shared substantive topic, SDG 17 policy discourse concerns coordinating and resourcing implementation itself, a conceptual space that technical AI research abstracts are not built to occupy.

2.2 A Three-Level Framework of Research-Policy Alignment

This paper proposes that research-policy alignment be understood as a three-level construct rather than a binary property of shared labels:

Level 1: lexical or bibliometric correspondence. The dominant mode of existing SDG-mapping work: assigning texts to goals through keyword dictionaries, Boolean queries, or citation and mention counts (Armitage et al., 2020; Kashnitsky et al., 2024). Necessary but weak: shared vocabulary does not guarantee shared meaning (Bornmann et al., 2016), and keyword methods are brittle to vocabulary and database choices (Armitage et al., 2020).

Level 2: observed textual-semantic proximity in embedding space. Sentence embeddings locate texts in a shared semantic space where proximity reflects distributional patterns of co-occurrence (Harris, 1954; Reimers & Gurevych, 2019), and encoder-based SDG association has been benchmarked as a strong baseline (Gjorgjevikj et al., 2025). This level detects divergence that keyword methods miss, while remaining a proxy rather than a definitive conclusion: embedding distance mixes modality, abstraction, institutional stance, and theme (Biber, 1988), and the research-policy contrast is primarily a register contrast (Biber & Conrad, 2009). The present study is positioned at this level.

Level 3: substantive comparison of implied problems, actors, and interventions. Whether the two discourses address the same problems, involve the same actors, and propose the same interventions (Cash et al., 2003; Guston, 2001; Haas, 1992). Because this requires interpreting causal beliefs, legitimacy, and institutional mechanism (Caplan, 1979; Haas, 1992), it lies beyond unsupervised text analysis and is left to qualitative follow-up.

Advancing from Level 1 to Level 2 is the study's concrete methodological contribution; Level 3 names the boundary it deliberately does not cross. Figure 1 illustrates this decomposition.

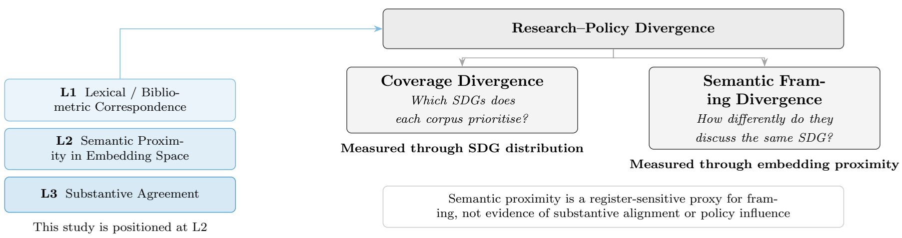

Figure 1. Conceptual framework: research–policy divergence separated into coverage and semantic framing dimensions, positioned within the three-level alignment construct.

2.3 Research-Policy Alignment in AI and Sustainability

A growing literature documents structural tensions between what AI research produces and what sustainability policy requires: corporate AI research concentrates on pre-deployment technical problems, creating a gap between research priorities and the deployment impacts governance must address (Strauss et al., 2025); global AI metrics emphasise technical and economic performance while policy discourse stresses ethical and social dimensions aligned with SDG concerns (Sioumalas-Christodoulou & Tympas, 2025); and research and policy communities can share trustworthy-AI terminology while differing in term salience and focal concerns (Toney-Wails et al., 2024). The pattern extends beyond AI: in conservation science, 59% of policy-prioritised research topics appear in fewer than 15 papers (McCarthy et al., 2026).

Adjacent scientometric work measures research-policy connection through policy-document mentions rather than semantic framing: only 1.2% of 191,276 climate-change publications have at least one policy mention (Bornmann et al., 2016). Mentions are a boundary marker of one narrow form of policy uptake; they do not show how research is discussed or framed inside policy discourse.

Theory explains why such divergence is the default rather than a failure. Knowledge systems support action when they produce information that is simultaneously salient, credible, and legitimate (Cash et al., 2003). The epistemic-communities framework explains why shared framing matters for policy coordination under uncertainty: expert communities influence policy by articulating cause–effect relationships and framing issues for collective debate (Haas, 1992). Haas's concept is broader than semantic proximity, involving shared normative and causal beliefs, validity criteria, and a common policy enterprise; the present study therefore does not operationalise an epistemic community directly, but measures only the weaker condition of observed semantic proximity. Even this weaker condition is only a possible precondition for knowledge transfer: semantic proximity cannot show the existence, strength, or influence of an expert community (Cross, 2013). The two-communities theory holds that researchers and policy makers live in separate worlds with different values, reward systems, and languages, so divergence is the default state of research-policy interaction (Caplan, 1979).

The benchmark itself is not neutral. The SDG framework is a politically negotiated instrument with documented limitations: goals can create incentives to over-focus on target achievement (Fukuda-Parr, 2016), summit documents remain state-centric and avoid causal responsibility (Bexell & Jönsson, 2017), and Goal 8's 3% annual growth target is empirically incompatible with the resource-use and emissions goals (Hickel, 2019). Reported SDG engagement is frequently superficial, so reported alignment should not be conflated with substantive commitment (Heilinger et al., 2024; Heras-Saizarbitoria et al., 2022). To the extent that research follows Mode 1 knowledge production (problems set within academic disciplines) rather than Mode 2 (problems arising in contexts of application) (Gibbons et al., 1994), divergence is not uniformly problematic: where research explores possibilities that policy discourse has not yet absorbed, distance may reflect productive exploration rather than failure of alignment.

2.4 Semantic Methods: Rationale and Cautions

The distributional hypothesis, that meaning can be described through patterns of linguistic co-occurrence (Harris, 1954), underlies the use of Sentence-BERT (Reimers & Gurevych, 2019) and related transformer encoders for large-scale semantic comparison. Applying them to compare two distinct discourse corpora, research abstracts and policy documents, is a novel usage rather than a directly benchmarked one, and the interpretation of distance remains descriptive rather than causal. Gjorgjevikj et al. (2025) benchmarked sentence encoders specifically for SDG association tasks, finding that general-purpose models from the sentence-transformers family perform among the strongest as baselines, and that domain-specific fine-tuning further improves predictive performance while reducing sensitivity to description length. Rodriguez et al. (2023)'s conText framework provides a computational-social-science precedent for comparing meaning across contexts. They adopt a narrow, "structuralist" view (in the sense of Harris (1954)) in which meaning arises from word co-occurrences alone, and explicitly make "no claims that the distributions per se have causal effects on human understandings of terms" (Rodriguez et al., 2023, p. 6); when they speak of meaning differing across groups, they mean the distribution of co-occurring words within a fixed window is different, in a "thin sense" (Rodriguez et al., 2023, p. 6). This study follows that same descriptive premise: semantic proximity is a distributional measurement, not evidence of substantive framing alignment. The implication is that embedding distance mixes topic, register, and contextual convention into a single number, motivating the decomposition into these components (Section 3.8). McCarthy et al. (2026) supply a closer empirical precedent, showing with a multi-target SciBERT classifier that research–policy coverage gaps are measurable with transformer encoders outside the AI–SDG setting; this study extends that logic from a single conservation domain to the full SDG landscape, and from coverage to within-SDG semantic framing.

An older corpus-linguistic tradition adds a further caution. Biber's multidimensional (MD) analysis identifies six co-varying dimensions of linguistic variation across 23 spoken and written genres, derived from factor analysis of 67 linguistic features (Biber, 1988). Each dimension captures a continuum of co-occurring linguistic features — for example, Dimension 1 contrasts involved, interactive discourse (first/second person pronouns, contractions, private verbs) with informational, edited prose (nouns, prepositions, attributive adjectives) — rather than a binary speech/writing split (Biber, 1988; Biber & Conrad, 2009). Register is therefore a multi-layered property: two texts can differ on some dimensions and converge on others. The INLP procedure in this study removes a *linear* subspace from the embedding, which is a projection onto one or more directions in the 768-dimensional space; this is best understood as capturing the dominant linear component of Biber's continuous register space, not as a claim that register is binary or that a single projection exhausts it. For the present study this matters because research abstracts and policy documents may differ not only in substantive topic, but also in abstraction, modality, institutional stance, and rhetorical function; embedding distance is therefore treated as observed textual-semantic proximity, not a clean measure of topic alone.

2.5 Research Gap

To the best of my knowledge, the literature reviewed above has not produced: (i) a systematic comparison of how academic AI-for-sustainability research and institutional SDG policy discourse differ in SDG coverage and within-SDG semantic framing; prior work measures research coverage, citation patterns, or policy-document uptake, but does not compare the two corpora directly on these dimensions; or (ii) an examination of whether coverage gaps and semantic gaps are positively related, weakly related, or inversely related. These are the contributions of the present study, and they answer the central research question set out in the Introduction. Together, they provide a systematic framework for distinguishing research-policy coverage from research-policy semantic proximity within a shared SDG representation. This framework is deliberately diagnostic: it maps where observed divergence is visible, without classifying whether any given gap reflects productive or problematic disconnection; the latter would require qualitative follow-up beyond the scope of embedding analysis.

3 Methodology

3.1 Research Corpus

This study uses only publicly available text data and code. It does not involve human participants or personal data. No ethical approval was required.

The research corpus consists of 2,536,771 English-language research abstracts retrieved from the OpenAlex API (Priem et al., 2022), covering publications between 2018 and 2025. Retrieval combined OpenAlex's native SDG work filter with four high-precision AI search terms (`artificial intelligence`, `machine learning`, `deep learning`, `neural network`), giving 68 queries (17 SDGs $\times$ 4 terms). The term set follows established practice for delimiting AI corpora (Hellwig et al., 2019; OECD, 2024); because the aim is macro-level semantic alignment rather than exhaustive publication counting, precision is prioritised at the retrieval stage to keep non-AI noise out of the embedding space. Complete query parameters are documented in the Supplementary Methodology (Appendix A).

A direct test of this scope condition re-ran retrieval using field-of-study membership for the same two AI fields (OECD, 2024) and re-scored the resulting 100,000-paper corpus through the identical pipeline. The coverage and semantic-gap rankings are stable under this alternative retrieval strategy (Section 4.4; Appendix H, Table 13), confirming that the reported divergence patterns are not an artefact of the four-term query choice.

An important asymmetry follows: the research corpus is a tight intersection of AI and SDG content (AI ∩ SDG), while the policy corpus (Section 3.2) is a broader union covering institutional SDG discourse and AI governance. Between-SDG rankings are conditional on this asymmetric construction; following Armitage et al. (2020), the retrieval universe is a method-dependent corpus boundary, not a neutral population of all SDG-relevant AI research. After deduplication and removal of records lacking usable abstracts, 2,536,771 abstracts were retained and tokenised into 3,105,144 segments (1.224 segments per abstract on average; 17.18% exceed the 374-token encoder window and split into multiple segments). A small number of records fail the 20-word segment minimum and are excluded: 2,543,698 records survive preprocessing, of which 6,927 are dropped at segmentation because their combined text is too short (mostly very short abstracts, plus a handful of mixed-language records whose English title is followed by a non-English abstract). This is a documented scope boundary, not an error, and the dropped share is below 0.3%. No balancing by SDG or citation count is applied, because trimming would distort the coverage profile the study aims to measure.

3.2 Policy Corpus

The policy corpus contains 40,597 text segments from three collections, representing 6,367 unique source documents. It should be read as mixed *international institutional SDG policy discourse* rather than the full universe of AI sustainability policy:

1. **Curated AI and SDG policy documents** (13,123 segments, 94 documents): UN, OECD, EU, UK, and AU AI governance frameworks; IPCC assessment reports; SDSN sustainable development reports; and national AI strategies. These are AI-governance instruments by construction, forming the innovation-facing slice of the corpus.
2. **SDGi Voluntary National Review and Voluntary Local Review corpus** (SDGi, 2024) (21,346 segments, 4,225 documents): National and local self-assessments submitted to the UN High-Level Political Forum. The creators describe it as the most comprehensive multilingual collection of SDG-labelled texts to date, spanning all 17 goals and centred on SDG implementation rather than any single sector.
3. **UN General Debate Corpus** (Harvard Dataverse, 2024) (6,128 segments, 2,048 documents): SDG-relevant passages from General Debate speeches, sessions 70–80 (2015–2024). Its creators analyse SDG coverage directly and report climate action (SDG 13), peace and governance (SDG 16), and partnerships (SDG 17) among the most prominent themes, an orientation the analysis below also observes.

After cross-source deduplication by exact text match (SDGi retained first), 40,597 unique segments remained. The three collections differ in genre and scope: the curated subset is AI-governance-focused, while SDGi and UNGDC reflect broader institutional SDG discourse, so the full-corpus distribution should be read accordingly; the curated sub-corpus (94 documents) is the more relevant reference for AI-governance-specific interpretation. Appendix I.6 reports a source-family sensitivity check. Segmentation is token-aware (Section 3.4); full algorithmic details are in Appendix A.2.

3.3 Embedding Model and Normalisation

All texts were encoded with `all-mpnet-base-v2` (Reimers & Gurevych, 2019; Sentence-Transformers, 2021) from the sentence-transformers library, producing 768-dimensional dense vectors. This pre-trained model requires no fine-tuning on the data under analysis, so the embedding space is a stable, pre-existing coordinate system. Encoder choice is a principal robustness axis: the gap rankings are re-computed under two alternative encoders (a general-purpose model and a domain-specialised scientific model, both 768-dimensional), and the cross-sensitivity results — together with assignment-method, policy-source, and segment-cap variations — are reported in Section 4.4 (Tables 3, 14) and Appendix I. Embeddings were L2-normalised to unit vectors so that cosine similarity equals their dot product.

3.4 Token-Aware Segmentation

All source documents are segmented at sentence boundaries using the embedding model's own tokenizer to enforce a hard token budget, guaranteeing that every segment fits within the encoder's context window. The procedure applies identically to research abstracts, policy documents, and reference corpora: NLTK sentence-boundary detection is followed by greedy accumulation of sentences up to a token budget of `max_seq_length` -10 (for the canonical MPNet encoder, $384 - 10 = 374$ tokens). The 10-token safety margin was verified empirically to reduce post-embedding truncation to near zero (Appendix A.3). Segments falling below 20 words are discarded; a single sentence exceeding the token budget is kept whole because mid-sentence splitting is not permissible. Cross-source deduplication by exact text match (SDGi retained first) is applied after segmentation.

For the research–policy comparison this design has two consequences. First, research abstracts (mean $\sim$200 words) mostly fit within the 374-token budget, but 17.18% exceed it and split into multiple segments; the research-side text unit is therefore the abstract, whose segments are aggregated to a single document vector before assignment, not the individual segment. Second, policy documents of variable length are split into multiple segments of comparable size to research abstracts, making the two corpora directly comparable in embedding space. The per-document segment cap of $K = 50$ applied when computing semantic-gap centroids (Section 3.7) is a separate, additional restraint on document-level dominance within a single SDG cluster; it does not affect the segmentation itself. Full algorithmic details are in Appendix A.2.

3.5 Supervised Reference Classifier

The canonical instrument is a supervised multinomial logistic regression (LR) classifier trained on pooled single-label reference data from five independent annotated corpora:

- **OSDG Community Dataset** (Pukelis et al., 2022) (30,532 texts, SDGs 1–16): crowdsourced excerpts from reports, policy documents, and abstracts, each validated by multiple volunteers.
- **SDG Classification Benchmark** (SDG Classification Expert Group, 2025) (281 expert-verified texts, SDGs 1–17): short text snippets annotated by multiple domain experts with consensus resolution.
- **IISD SDG Knowledge Hub** (Wulff et al., 2023) (6,122 texts, SDGs 1–17): IISD journalism and policy commentary on 2030 Agenda implementation.
- **SDGi Corpus** (SDGi, 2024) (20,267 texts, SDGs 1–17): government Voluntary National and Local Reviews submitted to the UN.
- **Aurora dataset** (Vanderfeesten et al., 2020) (4,971 expert-validated texts, SDGs 1–17): academic abstracts validated by domain researchers.

All five corpora supply human- or expert-validated labels rather than machine-generated ones, and they differ in provenance: OSDG is a crowdsourced volunteer-confirmed collection, SDGi compiles government reviews, the Knowledge Hub is journalism, and Aurora is academic expert-validated text. The mix broadens the label base, but OSDG alone contributes about 49% of labelled texts and draws on UN-adjacent documents, so the reference data remain weighted toward policy vocabulary. No high-quality graded relevance sets exist for SDG classification, making this combined, multi-source collection the best available resource (Kashnitsky et al., 2024).

The pooled reference data (n=62,173 after single-label filtering) were split into training and held-out test sets (n=52,835 train, n=9,338 test) by stratified document-grouped splitting, keeping all texts from the same source document in one fold to enforce strict separation between training and evaluation. The champion LR ($C = 3.0$, L2 penalty, `class_weight=None`, solver `lbfgs`) was selected by 5-fold GroupKFold cross-validation on the training pool and reaches macro-$F_1 = 0.816$ on the held-out test set (Section 4.1). The grid-search protocol, hyperparameter rationale, and the MLP robustness comparison are reported in Appendix E. The validation task does not evaluate a rejection option for non-SDG texts; this is acceptable because the research corpus is first restricted to an

OpenAlex AI-and-SDG universe, so the classifier answers a relative question among the 17 SDGs.

A text is assigned to the SDG with the highest predicted probability from the trained LR classifier (argmax over 17 classes). Every assignment is traceable to 768 signed logistic-regression coefficients per SDG class, a transparent decision boundary with no hidden layers, no dropout, and no random-seed sensitivity. Macro-F_1, the unweighted mean of per-SDG scores, is the primary validation metric (Section 4.1); LR also provides calibrated per-class probabilities used as a soft-assignment comparison against the canonical hard argmax.

Because the classifier is trained on pooled reference data that are themselves policy-adjacent, the scoring instrument behaves differently on the two corpora. Policy segments receive systematically higher LR predicted probabilities than research abstracts (mean 0.749 vs 0.627, a 0.122 gap). The gap has two causes the data cannot separate: a genre artefact (the reference data skew toward policy vocabulary) or a substantive difference (policy texts address SDG implementation more directly). Either way the asymmetry does not threaten the findings: hard argmax assignment uses only within-corpus ranking, not absolute probability levels, so the directional gap does not reorder goals. It is not an SDGi circularity effect: the training set and scored policy corpus are disjoint by construction. The combined PCA landscape visualises this asymmetry (Figure 4a, Results): the policy cloud hugs the 17 reference-classifier centroids more tightly than the research cloud. The projection is descriptive only (the first two components explain 9.2% and 3.7% of variance); all reported cosine distances remain in the original 768-dimensional space.

3.6 Coverage Gap Analysis

For each corpus item, the predicted SDG is the LR argmax (hard assignment, Section 3.5); the coverage profile is the proportion of items assigned to each SDG.

The policy corpus is not a flat set of documents: its 40,597 segments are grouped into 6,367 source documents of very different lengths (Section 3.2). A segment-weighted profile would let a handful of long reports dominate the count, so the analysis instead computes a *document-weighted* profile in which every source document counts equally regardless of length. Concretely: (1) the segment score vectors of each document are averaged into one document-level vector; (2) that vector is hard-assigned to its top SDG; (3) the coverage profile is the share of documents, not segments, assigned to each SDG. The research corpus uses the same logic at the paper level: each of the 2,536,771 abstracts is represented by the mean of its segments, hard-assigned as a single document unit, so the research-side text unit is the abstract and the two corpora are weighted identically. Document-weighted profiles are the canonical representation; segment-weighted profiles are retained as a diagnostic comparison.

The coverage gap for SDG j is defined as:

$$\text{CoverageGap}_j = \left|\text{ResearchCoverage}_j - \text{PolicyCoverage}_j^{\text{docwt}}\right|$$

The signed version of the same quantity is:

$$\text{SignedCoverageGap}_j = \text{ResearchCoverage}_j - \text{PolicyCoverage}_j^{\text{docwt}}$$

A positive value means research dominates SDG j; a negative value means policy dominates. These two definitions, together with research coverage and policy coverage, are the four predictors reported in Table 3.

3.7 Semantic Gap Analysis

The semantic gap measures within-SDG divergence: given that both corpora have texts assigned to SDG j, how differently do they discuss it? For each SDG j, the analysis collects all assigned research papers and all assigned policy segments (capped at 50 per source document to prevent single-document dominance; SDSN reports alone contribute up to 3,179 segments otherwise), computes the L2-normalised mean embedding of each cluster ($\mathbf{c}_j^{\text{res}}$, $\mathbf{c}_j^{\text{pol}}$), and defines:

$$\text{SemanticGap}_j = 1 - (\mathbf{c}_j^{\text{res}} \cdot \mathbf{c}_j^{\text{pol}})$$

A gap of 0 indicates perfectly overlapping semantic directions; a gap approaching 1 indicates orthogonal framing.

The *per-document segment cap* of $K = 50$ (approximately 7,500 words) uses seeded random sampling, hence reproducible, and sits well above the median of roughly 14 segments per document. The sensitivity of the gap ranking to the per-document segment cap was tested by repeating the computation at a cap of 20 and with no cap (Table 13, Section I.2).

3.8 Register Adjustment via Iterative Nullspace Projection (INLP)

The semantic gap is not a clean measure of substantive disagreement. Following Biber and Conrad's distinction between register, genre, and style (Biber & Conrad, 2009), I treat the research–policy contrast primarily as a register contrast: the two corpora differ in situation of use, audience, and communicative purpose, not merely in topic. Embedding distance therefore mixes genuine within-SDG substantive divergence with register divergence in abstraction, modality, and institutional stance (Biber, 1988). To separate these components, I decompose the raw centroid distance into a register-adjusted (topic) component and a register component using Iterative Nullspace Projection (INLP; Ravfogel et al. (2020)). After register removal, the adjusted gap is the primary estimate of within-SDG topic divergence, while the residual register component captures the removed register signal (Table 2).

Objective. Following Ravfogel et al. (2020), let $\mathbf{x}_i \in \mathbb{R}^d$ be an embedding and $z_i \in \{\text{research}, \text{policy}\}$ its corpus label. The goal is to learn a linear guarding function $g : \mathbb{R}^d \to \mathbb{R}^d$ such that no linear classifier can predict z_i from $g(\mathbf{x}_i)$ at above majority-class accuracy. After K iterations the composite projector is $P = P_{N(W_K)} \cdots P_{N(W_1)}$, with $g(\mathbf{x}) = P\mathbf{x}$. Each W_k is a logistic-regression weight matrix whose rowspace carries the linear signal used for prediction; projecting onto the nullspace $N(W_k) = \{\mathbf{x} : W_k\mathbf{x} = \mathbf{0}\}$ zeroes out exactly those components, rendering W_k useless on the projected data. The orthogonal nullspace projection is the least harmful linear operation for this purpose: among all maximum-rank projections onto $N(W_k)$, it preserves distances optimally because the image under an orthogonal projection is by definition the closest point in the subspace.

SDG-stratified procedure. Vanilla INLP removes a binary attribute (e.g. gender). The present adaptation must remove *register* — a corpus-level property that is confounded with SDG topic in the raw embeddings. The key modification is SDG stratification: at each iteration, exactly 1000 research and 1000 policy embeddings are drawn from each of the 17 SDGs, preventing the classifier from learning topic membership as a proxy for register. The procedure proceeds as follows at each iteration $k = 1, \ldots, K$:

1. **Stratified sample.** Draw $n_{\text{target}} = 1000$ texts per SDG per corpus (global-minimum rule: $n_{\text{target}} = \min_j \min(|X_j^{\text{res}}|, |X_j^{\text{pol}}|)$; under the frozen snapshot, $n_{\text{target}} = 1207$ for MPNet). Form the 34-class stratification key $z_{\text{strat}} = 2 \times \text{sdg\_labels} + y$, ensuring balanced corpus × topic combinations.
2. **Project existing directions.** Subtract from each sample embedding the components along all previously learned directions: $\tilde{\mathbf{x}} = \mathbf{x} - \sum_{j<k}(\mathbf{g}_j \cdot \mathbf{x})\,\mathbf{g}_j$.
3. **Train classifier.** Fit L2-regularised logistic regression ($C = 1.0$, solver: lbfgs, max_iter: 1000) to predict corpus identity y from the projected embeddings $\tilde{\mathbf{x}}$. Split 15% held-out, stratified by z_{strat}.
4. **Extract direction.** Normalise the weight vector: $\mathbf{g}_k = \mathbf{w}_k/\|\mathbf{w}_k\|$. Orthogonalise against all prior directions: $\mathbf{g}_k \leftarrow \mathbf{g}_k - \sum_{j<k}(\mathbf{g}_k \cdot \mathbf{g}_j)\,\mathbf{g}_j$. If $\|\mathbf{g}_k\| < 10^{-8}$ (direction collapse), stop.
5. **Stopping criterion.** Stop if held-out accuracy ≤ 0.5 (the majority-class baseline for balanced binary classification). Otherwise append $\mathbf{g}_k$ and continue.

The maximum number of iterations is $K = 200$. The final adjusted embedding is

$\mathbf{x}' = \mathbf{x} - \sum_{k=1}^{K} (\mathbf{g}_k \cdot \mathbf{x})\, \mathbf{g}_k$, and the adjusted gap uses centroids computed on these projected embeddings. The stopping criterion is held-out accuracy ≤ 0.5 (majority-class baseline for balanced binary classification); convergence behaviour is reported in Appendix F.1 (Table 9).

Identification argument. The stratification design is the critical structural precondition. Because every iteration trains on equal numbers of research and policy embeddings from each SDG, the classifier cannot learn which SDG a document belongs to: every SDG is balanced across classes. The only remaining signal available for discrimination is the corpus-level register difference — the way research papers sound versus the way policy documents sound, within the same substantive topic. What gets removed at each iteration is therefore precisely the register component: modality differences (academic prose versus policy prose), audience conventions (peer reviewers versus policymakers), institutional stance (hedged claims versus prescriptive language), and structural patterns (methods sections versus action items). What survives is the semantic content difference between AI research and SDG policy for each specific SDG.

This identification argument is structural, not empirical: the stratification design *ensures* that no linear direction can simultaneously encode SDG topic and corpus identity, because every SDG is balanced across the two corpora at every iteration. What remains linearly decodable is therefore the corpus-level register difference. The removed subspace is interpreted as register in the sense of Biber and Conrad (2009): a variety associated with a particular situation of use, including communicative purpose, audience, and production circumstances. I evaluate this interpretation against independent linguistic markers (hedge-word density, passive-voice frequency, mean sentence length, and related Biber-style features (Biber, 1988)) in Appendix G. That validation finds a substantial reduction of the corpus-level signal after register removal, but not its complete elimination, and no robust evidence that topic is being systematically removed; the residual limitations are discussed in the appendix.

Geometric effect. The effect of INLP register removal is shown in Results (Figure 4): the raw embedding space separates research and policy clouds, while the adjusted space merges them, confirming that the separation is driven by register rather than topic. The effect is encoder-dependent — SciBERT already fuses the clouds without INLP — so this figure is MPNet-specific.

With the projector learned, the semantic-gap analysis of Section 3.7 is recomputed twice: once on the adjusted embeddings, yielding the within-SDG *adjusted gap* (the primary estimate of topic divergence), and once on the removed component $(\mathbf{x} - \mathbf{x}')$, yielding the *register gap* (the amount of the raw gap attributable to register). The raw, adjusted, and register gaps together make up the decomposition reported in Results (Table 2).

3.9 Hypothesis and Analysis Plan: Coverage and Semantic Gaps Interaction

The first hypothesis (H1) decomposes into four sub-hypotheses, each correlating a distinct coverage predictor with the within-SDG semantic gap across the 17 SDGs: H1a, coverage gap $\leftrightarrow$ semantic gap; H1b, research–policy dominance $\leftrightarrow$ semantic gap; H1c, research coverage $\leftrightarrow$ semantic gap; and H1d, policy coverage $\leftrightarrow$ semantic gap. H1a is the motivating hypothesis; H1b–H1d examine complementary coverage-structure predictions. I compute Pearson r and Spearman ρ between each predictor and semantic-gap scores; SDG 4 is treated as artefact-affected and examined separately (Appendix B.1). Given $n = 17$, the tests have limited power to detect anything but a large linear effect; this is a scope condition.

3.10 Methodology Summary

Figure 2 summarises the end-to-end methodology. All three tracks share preprocessing (cleaning, English filtering) and token-aware segmentation to the encoder's token budget with a 20-word minimum (Section 3.4). The reference track pools 62,173 single-label texts (52,835 train, 9,338 test) to train the LR classifier ($C = 3.0$, L2, macro-$F_1 = 0.816$); the research and policy tracks segment 2,536,771 abstracts into 3,105,144 segments and 40,597 policy segments respectively. All tracks are embedded with `all-mpnet-base-v2` (768-D, L2-normalised) and scored via argmax over 17 SDG classes.

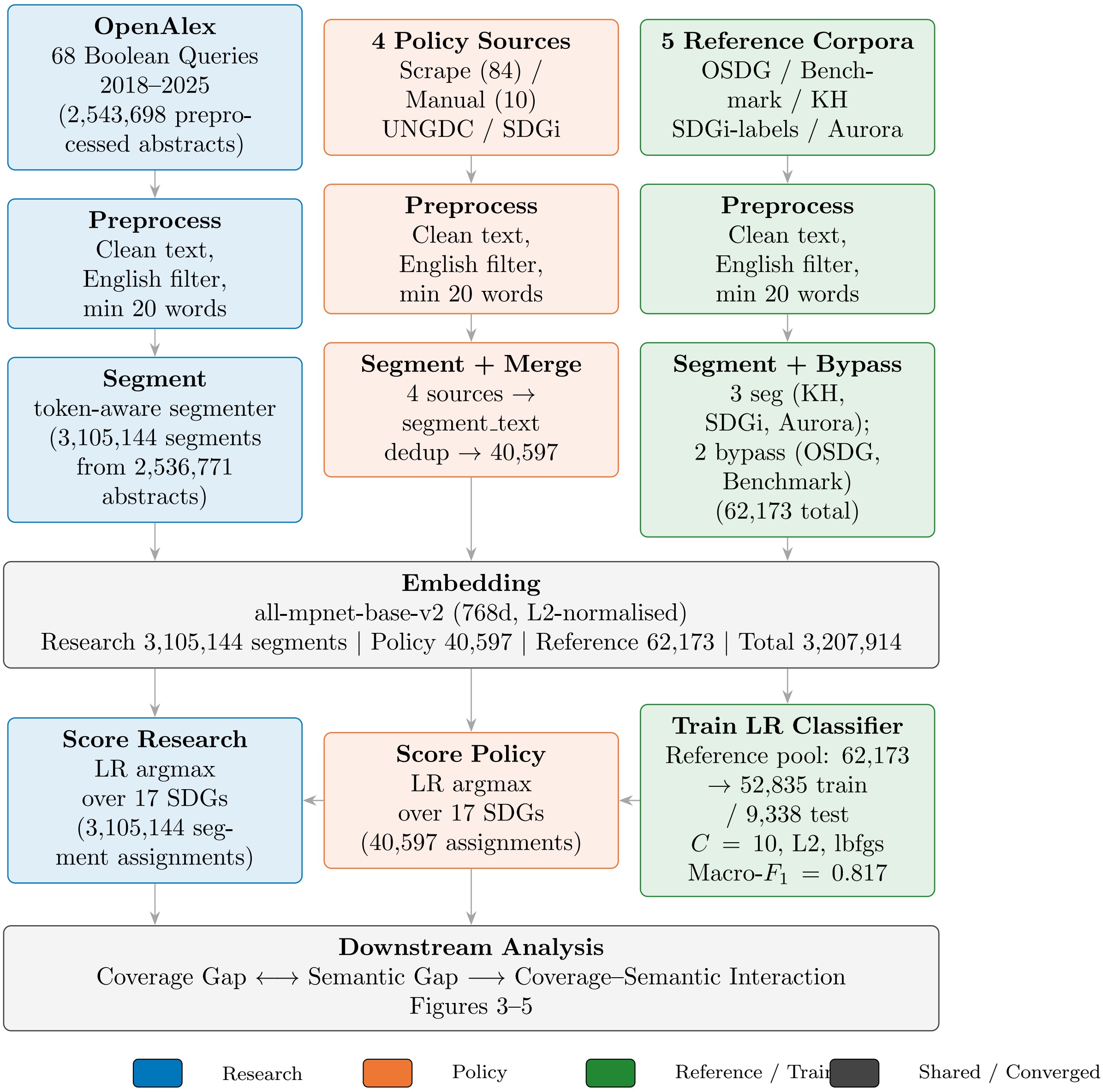

Figure 2. End-to-end methodology pipeline in three parallel data tracks.

4 Results

This section reports the validation scores, coverage and semantic gaps, H1 correlation tests, and the robustness of those rankings, without interpretation; interpretation, limits, and implications follow in the Discussion (Section 5). All rank-robustness tables in this paper follow one convention: the adjusted (register-removed) ranking is Panel (a) and the raw (naive) ranking is Panel (b), with the tested measurement dimension nested inside each panel.

4.1 Supervised Reference Classifier and Coverage Gap

The LR classifier was evaluated on the held-out test set (n=9,338, document-grouped to prevent source leakage). It reaches macro-F1 = 0.816 and micro-F1 = 0.8306, far above the 1/17 random baseline (0.059). Per-class F1 values from the test set are reported in Table 1.

Table 1. LR test-set F1 scores (n=9,338).

SDG	LR test F1
SDG 1 (No Poverty)	0.737
SDG 2 (Zero Hunger)	0.830
SDG 3 (Good Health)	0.903
SDG 4 (Quality Education)	0.889
SDG 5 (Gender Equality)	0.881
SDG 6 (Clean Water)	0.879
SDG 7 (Clean Energy)	0.861
SDG 8 (Decent Work)	0.716
SDG 9 (Industry & Infra.)	0.770
SDG 10 (Reduced Inequalities)	0.610
SDG 11 (Sustainable Cities)	0.762
SDG 12 (Responsible Cons.)	0.783
SDG 13 (Climate Action)	0.831
SDG 14 (Life Below Water)	0.892
SDG 15 (Life on Land)	0.865
SDG 16 (Peace & Justice)	0.879
SDG 17 (Partnerships)	0.788
Macro-F1 (SDGs 1–17)	**0.816**

Notes: LR test F1 on the held-out pooled test set (all reference sources). The table shows the canonical 17-way classifier used throughout the main analysis.

The LR achieves consistently strong per-SDG performance: SDG 3 (F1 = 0.903), SDG 14 (0.892), and SDG 4 (0.889) are the most confidently classified (each above 0.88). The weakest classes are SDG 10 (Reduced Inequalities, F1 = 0.610) and SDG 8 (Decent Work, F1 = 0.716), consistent with their semantic breadth and the overlapping cluster structure of adjacent goals (Kashnitsky et al., 2024) (Appendix B.2, Figure 7). Most goals score between 0.76 and 0.89, indicating reliable discrimination across the 17-class space. SDG 17 (F1 = 0.788) benefits most from the supervised classifier: the decision boundary separates partnership discourse more cleanly than the centroid average.

The MLP robustness check (Appendix E) achieves a comparable test macro-F1 (MLP: 0.826; LR: 0.816), confirming that architecture choice is not the binding constraint. LR provides 768 signed coefficients per class, each directly attributable to a semantic dimension, while MLP's hidden layers diffuse this information across non-linear transformations.

In summary, these scores mark a ceiling for the embedding-plus-linear-classifier approach; the comparable MLP result confirms the ceiling is not an artefact of linear separability assumptions.

Figure 3 presents the document-weighted policy coverage profile alongside the research coverage profile. The two profiles are markedly different.

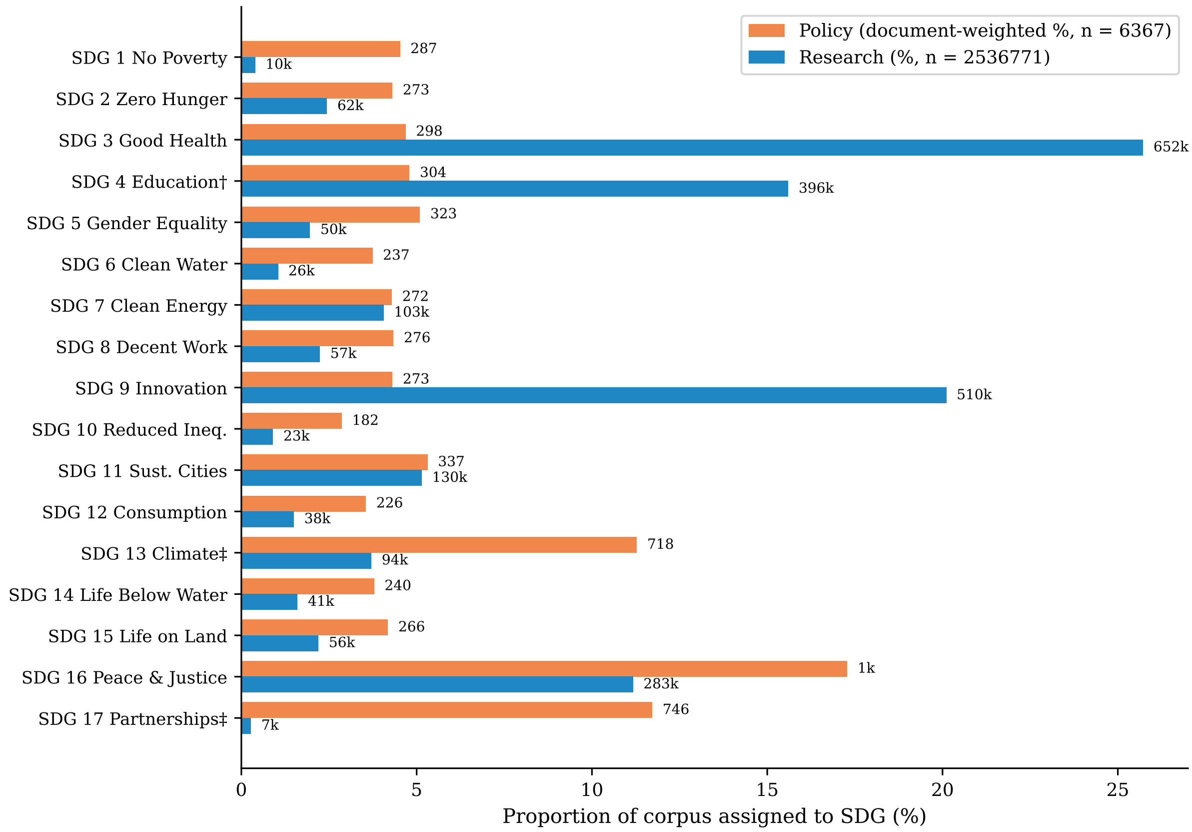

Figure 3. Coverage profiles by SDG. SDG 4 is flagged as artefact-affected due to ML-learning vocabulary overlap with education corpora (Appendix B.1).

Policy corpus. Three SDGs dominate the document-weighted profile of the full institutional policy corpus: SDG 16 (Peace, Justice and Strong Institutions, 17.3%), SDG 13 (Climate Action, 11.3%), and SDG 17 (Partnerships for the Goals, 11.7%), accounting for 40.3% of policy documents. This ordering is consistent with the literature review's account of SDG 17 as the cross-cutting means-of-implementation goal (Section 2, (Griggs et al., 2017; Le Blanc, 2015)). The SDG 16 lead is notable: under the supervised classifier, governance and justice discourse emerges as the most prevalent policy theme, ahead of both climate and partnerships. Appendix I.6 shows this full-corpus profile is source-family sensitive: the curated AI/SDG sub-corpus is much more innovation-heavy, while SDGi and UNGDC are more climate-, partnerships-, and governance-oriented.

Research corpus. The research profile is broader, but it still concentrates strongly on a smaller set of SDGs. The largest research shares are SDG 3 (Good Health, 25.7%), SDG 9 (Innovation, 20.1%), and SDG 4 (Quality Education, 15.6%), followed by SDG 16 and SDG 11 at 11.2% and 5.1% respectively. The SDG 3 lead corroborates the literature review, where SDG 3 is the one goal every major AI and sustainability bibliometric study finds dominant Bautista-Puig et al. (2021), Cowls et al. (2021), Nedungadi et al. (2024), and Singh et al. (2024). The SDG 4 share is the one point of disagreement: prior inventories do not rank SDG 4 among the leaders, so its third-place finish here is the same result flagged as likely inflated by machine-learning vocabulary (Appendix B.1). Even so, the research corpus clearly does not mirror the policy-side dominance of SDGs 16, 13, and 17.

Largest coverage mismatches. The five largest absolute coverage gaps are SDG 16 (0.061), SDG 3 (0.210), SDG 17 (0.114), SDG 13 (0.076), and SDG 9 (0.158). SDG 16 is the most asymmetric goal: policy discourse strongly emphasises governance and institutions while research barely touches it, followed by the familiar research-over-policy dominance in health (SDG 3).

Taken together, three observations support reading the coverage profiles as a real signal rather than a measurement artefact. First, the instrument reaches macro-F1 = 0.816 on the held-out benchmark (Section 4.1), far above the 1/17 random baseline. Second, the SDG 3 dominance reproduced here is consistent with four prior bibliometric studies that independently find SDG 3 leading AI-for-sustainability research (Section 2), even though those studies used keyword and citation methods rather than embeddings. Third, this is all while the classifier is trained on reference corpora that are themselves policy-adjacent (Section 3.5); that the ranking nonetheless converges with independent, methodologically unrelated literature is the strongest available evidence that the classifier anchors a meaningful semantic coordinate system, not a policy-specific artefact.

The coverage-gap ranking is stable across embedding architectures and retrieval strategy; the cross-encoder, cross-retrieval, cross-segment-cap, and cross-policy-source-family rankings are reported in Appendix I (Tables 16, 15, 13).

4.2 Semantic Gap after Register Removal

Panel (a) of Figure 4 shows the combined research–policy embedding space: the policy cloud hugs the 17 reference-classifier centroids more tightly than the research cloud. The projection is descriptive only (the first two components explain 9.2% and 3.7% of variance); all reported cosine distances remain in the original 768-dimensional space.

Figure 4 shows the geometric effect of INLP register removal: the raw embedding space (left) separates research and policy clouds, while the adjusted space (right) merges them, confirming that the separation is driven by register rather than topic. The effect is encoder-dependent — SciBERT already fuses the clouds without INLP — so this figure is MPNet-specific.

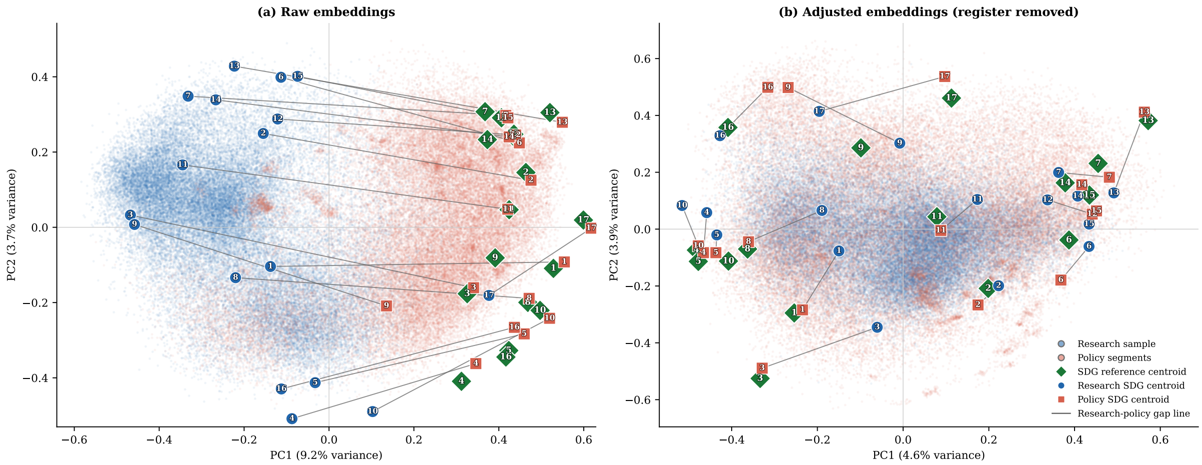

Figure 4. PCA before/after INLP register removal (MPNet).

Figure 5 shows the within-SDG semantic gap for all 17 SDGs, sorted by the adjusted (topic) gap after register removal. The adjusted gap ranges from 0.326 (SDG 17, widest) to 0.084 (SDG 15, narrowest), a span of 0.282 cosine units. The raw gap (naive baseline) ranges from 0.480 (SDG 13, widest) to 0.177 (SDG 17, narrowest), a span of 0.317. A large gap means research and policy use different words and concepts for the same goal; a small gap, that they frame it the same way. The adjusted ranking is stable when the segment cap is 20 per document or removed (Table 13), so the result does not depend on the sampling parameter. A second sampling concern is corpus asymmetry: research (2,536,771 abstracts) outweighs policy (6,367 documents) by roughly 398:1 in document units. I therefore re-derived the ranking on random research draws comparable in size to the policy corpus: across 100 independent 50,000-paper draws it reproduces the full-corpus ordering at Spearman $\rho = 0.986$ (SD 0.009), so the research-side over-representation does not drive which SDGs show large versus small gaps (Appendix D).

Largest semantic gaps (adjusted, primary). SDG 17 (Partnerships, 0.326), SDG 3 (Good Health, 0.319), SDG 16 (Peace, 0.242), SDG 1 (No Poverty, 0.191), and SDG 10 (Reduced Inequalities, 0.195) show the widest adjusted gaps — a mix of partnerships, health, governance, and inequality goals. Under the raw gap (naive baseline), the ordering shifts: SDG 13 (Climate Action, 0.480) leads, followed by SDGs 3, 11, 1, and 12. The register decomposition confirms that the raw and adjusted orderings differ because register uniformly inflates the raw gap, except for SDG 17 where register *increases* apparent similarity.

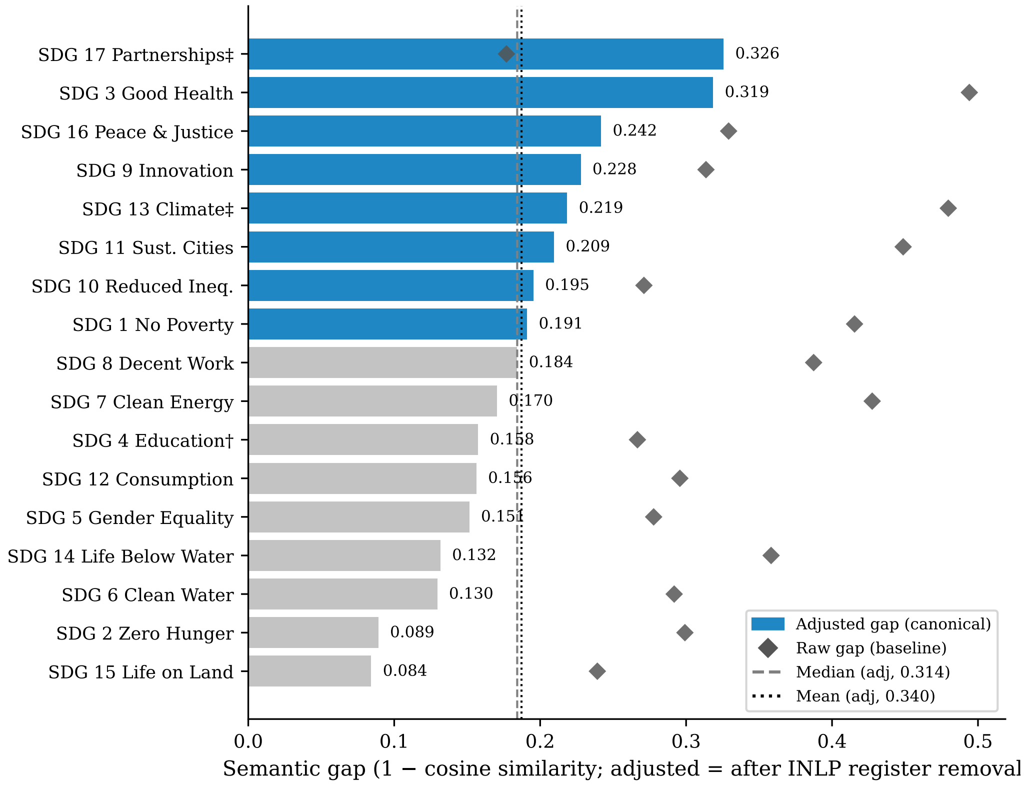

Figure 5. Within-SDG semantic gap by SDG, sorted by the adjusted (topic) gap after register removal.

Smallest semantic gaps (adjusted, primary). SDG 15 (Life on Land, 0.084), SDG 2 (Zero Hunger, 0.089), SDG 6 (Clean Water, 0.130), SDG 14 (Life Below Water, 0.132), and SDG 4 (Quality Education, 0.158) show the narrowest adjusted gaps, meaning research and policy frame these goals in similar terms after register removal. A narrow gap does not mean balanced attention: SDG 9 is still research-dominant, and SDG 4 carries the learning-vocabulary concern (Appendix B.1). Appendix C.1 shows example texts for high- and low-gap cases.

The semantic-gap ranking is stable across encoders, retrieval strategy, segment cap, and policy-source family; the corresponding rank-robustness tables are reported in Appendix I (Tables 13, 14).

4.3 Coverage–Semantic Interaction: A Possible Cancellation

Table 3 reports the H1a–H1d correlation tests between coverage predictors and the within-SDG semantic gap, computed across every encoder–classifier configuration. Each block is one hypothesis; each row is one config; the three columns are the Spearman ρ of the predictor against the raw gap, the adjusted (topic) gap, and the register component (raw − adjusted). Unlike the rank-robustness tables, raw, adjusted, and register are reported as adjacent columns here because the apparent cancellation *for each config* is the object of

Table 2. Register-topic decomposition of the semantic gap.

SDG	Description	Raw Gap	Adj. Gap	Reg. Comp.	Cov. Gap
SDG 1	No Poverty	0.415	0.191	+0.224	0.0413
SDG 2	Zero Hunger	0.299	0.089	+0.210	0.0187
SDG 3	Good Health and Well-Being	0.494	0.319	+0.176	0.2104
SDG 4	Quality Education	0.267	0.158	+0.109	0.1081
SDG 5	Gender Equality	0.278	0.151	+0.126	0.0314
SDG 6	Clean Water and Sanitation	0.292	0.130	+0.162	0.0269
SDG 7	Affordable and Clean Energy	0.428	0.170	+0.257	0.0023
SDG 8	Decent Work and Economic Growth	0.387	0.184	+0.203	0.0210
SDG 9	Industry, Innovation and Infrastructure	0.314	0.228	+0.086	0.1580
SDG 10	Reduced Inequalities	0.271	0.195	+0.076	0.0196
SDG 11	Sustainable Cities and Communities	0.449	0.209	+0.239	0.0017
SDG 12	Responsible Consumption and Production	0.296	0.156	+0.139	0.0206
SDG 13	Climate Action	0.480	0.219	+0.261	0.0757
SDG 14	Life Below Water	0.358	0.132	+0.227	0.0219
SDG 15	Life on Land	0.239	0.084	+0.155	0.0198
SDG 16	Peace, Justice and Strong Institutions	0.329	0.242	+0.087	0.0611
SDG 17	Partnerships for the Goals	0.177	0.326	-0.149	0.1145

Notes: Raw gap is the reference semantic-gap estimate; Adjusted gap is after INLP register removal (primary estimate; evaluated in Appendix G); Register component = raw − adjusted. Coverage correlations are Spearman ρ across 17 SDGs.

comparison.

The four predictors are research coverage, policy coverage, the absolute research–policy coverage gap, and the signed research–policy dominance. MLP configs reuse the INLP projection matrix learned on their encoder's LR classifier (the same G), and the Concept rows use the concept-retrieved research corpus (Appendix H) with the same policy corpus.

The evidence does not support H1a on the raw gap: across the 17 SDGs, coverage divergence and the raw semantic gap show no detectable association ($\rho = -0.012$, $p = 0.963$; Table 3, MPNet LR row). With only 17 SDG-level observations the design can reliably detect only very large correlations ($|\rho| \geq 0.63$ at 80% power, $\alpha = 0.05$)[1]; this is evidence against a strong monotonic relationship, not proof of independence. The raw gap masks a register–topic decomposition: the raw correlation is near-zero because the register component ($\rho = -0.390$) appears to cancel the topic signal ($\rho = 0.544$; Table 2). Table 3 shows this apparent cancellation replicates across every encoder–classifier config. Under the raw gap all four H1 hypotheses are null; after register removal the adjusted topic gap is positively associated with coverage divergence (H1a: MPNet LR $\rho = 0.544$, $p = 0.024$; positive in 8/9 configs, e.g. Concept LR $+0.434^{\dagger}$) and with policy coverage (H1d: positive in 9/9

[1]Minimal detectable correlation under the Fisher-z approximation: $\mathrm{MDE} = \tanh\left((z_{0.975} + z_{0.80})/\sqrt{n-3}\right) = \tanh(2.802/3.742) \approx 0.63$ for a Pearson test at $n = 17$; the rank-based Spearman tests reported here have marginally lower power, so this is the reference bound.

Table 3. H1a–H1d coverage-predictor vs semantic-gap Spearman correlations across encoder–classifier configs.

	Raw gap	Adj. gap	Register
H1a. Coverage gap ↔ Semantic gap			
MPNet LR	-0.012	+0.544*	-0.390
MPNet MLP	-0.135	+0.336	-0.370
MPNet ZS	-0.243	+0.245	-0.392
MiniLM LR	-0.243	+0.397	-0.463†
MiniLM MLP	-0.042	+0.505*	-0.272
SciBERT LR	-0.130	+0.221	-0.267
SciBERT MLP	-0.333	-0.034	-0.120
Concept LR	+0.108	+0.434†	-0.194
Concept MLP	-0.017	+0.181	-0.140
H1b. Policy–research dominance ↔ Semantic gap			
MPNet LR	+0.159	-0.059	+0.029
MPNet MLP	+0.020	-0.118	-0.034
MPNet ZS	-0.025	-0.377	+0.306
MiniLM LR	+0.404	-0.020	+0.429†
MiniLM MLP	+0.319	-0.066	+0.348
SciBERT LR	+0.549*	-0.118	+0.547*
SciBERT MLP	+0.336	-0.429†	+0.529*
Concept LR	+0.051	-0.029	-0.260
Concept MLP	-0.174	+0.093	-0.199
H1c. Research coverage ↔ Semantic gap			
MPNet LR	+0.449†	+0.267	+0.145
MPNet MLP	+0.262	+0.098	+0.088
MPNet ZS	-0.044	-0.301	+0.309
MiniLM LR	+0.444†	+0.262	+0.331
MiniLM MLP	+0.373	+0.275	+0.279
SciBERT LR	+0.605*	+0.466†	+0.020
SciBERT MLP	-0.005	-0.196	+0.225
Concept LR	+0.221	+0.017	-0.076
Concept MLP	-0.098	-0.010	-0.127
H1d. Policy coverage ↔ Semantic gap			
MPNet LR	+0.212	+0.589*	-0.009
MPNet MLP	+0.175	+0.428†	-0.007
MPNet ZS	-0.004	+0.277	-0.240
MiniLM LR	-0.015	+0.449†	-0.159
MiniLM MLP	+0.083	+0.456†	-0.093
SciBERT LR	-0.100	+0.605*	-0.686**
SciBERT MLP	-0.172	+0.395	-0.395
Concept LR	+0.369	+0.510*	+0.178
Concept MLP	+0.256	+0.258	+0.082

Notes: Raw gap = main semantic-gap estimate; Adj. gap = after INLP register removal (main reported estimate; evaluated in Appendix G); Register = raw − adjusted. Each row is an independent computation over 17 SDGs. ***$p < .001$, **$p < .01$, *$p < .05$, †$p < .10$.

configs, significant for MPNet LR and the Concept and SciBERT LR configs at $p < 0.05$, with MPNet MLP marginal at $p < 0.10$). H1b (signed dominance) and H1c (research coverage) do not show a consistent adjusted-direction signal. Coverage and framing are therefore potentially linked through topic, not independent.

A metric-sensitivity check repeats this grid with Pearson r on the raw magnitudes instead of Spearman ρ (Appendix I.5); the adjusted-gap association is unchanged for the supervised tracks across all four H1 hypotheses, and the only reversals are the already-flagged lower-trust SciBERT configs, confirming the rank-based reporting is not driving the result.

The SDG 4 lexical artefact (Appendix B.1) does not drive this result. A leave-one-out check with SDG 4 removed leaves the raw research–gap correlation null ($\rho = 0.606$, $p = 0.013$) and preserves the positive register-adjusted coverage-divergence association ($\rho = 0.597$, $p = 0.015$), so the artefact props up neither the raw null nor the adjusted association.

The corpus asymmetry does not manufacture this null. Because the research corpus is the AI $\cap$ SDG intersection while the policy corpus pools AI-specific and broad institutional SDG discourse, I re-ran the interaction test with the policy side restricted to the curated AI/SDG family, a symmetric AI/SDG-vs-AI/SDG comparison that matches the research corpus's scope. The cancellation pattern holds: research share remains uncorrelated with the gap ($\rho = -0.06$, $p = 0.830$), and the coverage-gap correlation does not reach significance either ($\rho = -0.22$, $p = 0.390$; Table 22, Appendix I.6).

Figure 6 shows each SDG's adjusted semantic gap against its coverage gap; open grey markers show the raw baseline. The most striking feature is the absence of a strong pattern: SDGs with large coverage gaps occupy the full range of adjusted semantic gaps (e.g. SDG 3 has both the largest coverage gap and the second-largest adjusted gap, whereas SDG 9 has the second-largest coverage gap but a near-median adjusted gap), and research attention coexists with both low- and high-gap cases (e.g. SDG 9 high-coverage near-median-gap, SDG 3 high-coverage high-gap). No SDG drives the weak association; the near-null H1a result therefore holds across configurations.

4.4 Robustness of the Gap Rankings

The gap rankings reported above are not equally stable across all measurement choices. This section states the cross-method conclusions once; the full rank-robustness tables are in Appendix I (Tables 13, 14, 15, 16), and the distributional-robustness tables in Appendix C.2.

Cross-sensitivity (policy source, segment cap, retrieval). Under the canonical MPNet–LR configuration, the within-SDG semantic-gap ranking is stable across policy source family (curated AI/SDG, SDGi, UNGDC), segment cap (20, none), and retrieval

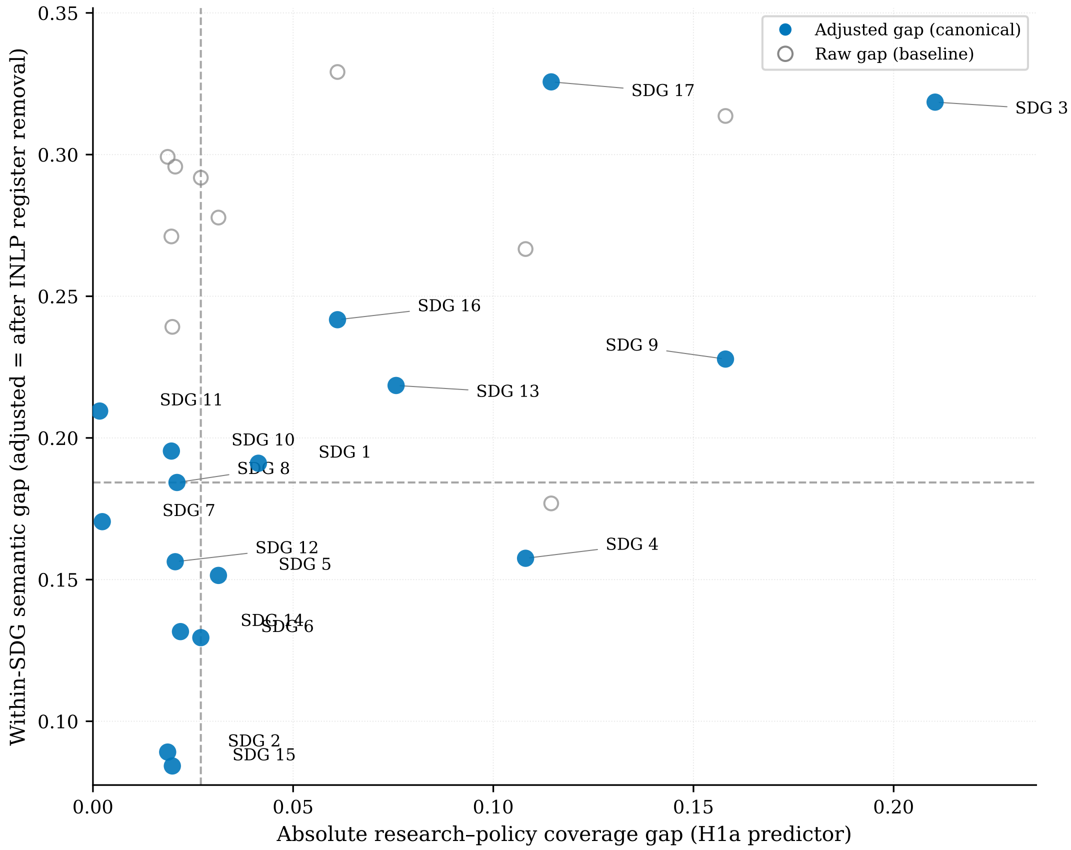

Figure 6. Coverage gap vs semantic gap, one point per SDG.

strategy (concept-based field-of-study): rank correlations with the canonical column are $\rho \geq 0.65$ throughout (Table 13). The same holds for the coverage gap (Appendix I, Table 16).

Encoder sensitivity. Coverage-gap ranks are highly stable across the three encoders (MPNet–LR vs. SciBERT–LR $\rho = 0.93$; vs. MiniLM–LR $\rho \approx 0.63$, Table 15). The semantic-gap *magnitude* is encoder-dependent: SciBERT gives a mean gap of 0.04 against MPNet's 0.34, because its scientific pretraining (Beltagy et al., 2019) fuses the two genres (Appendix F); but the divergence conclusion and the coverage-structure invariance hold across general-purpose and domain-specialised encoders alike (Table 14). SDG 13 leads and SDG 17 trails under both general-purpose encoders; the scientific-domain SciBERT compresses the magnitudes and, at its LR assignment, shifts the lead to SDG 3 and the trail to SDG 8.

Assignment-method sensitivity. Under the canonical encoder, LR, the MLP robustness check, and zero-shot nearest-centroid show varied agreement (MPNet LR vs. MLP $\rho = 0.85$, LR vs. zero-shot $\rho = 0.60$, Table 14). SDG 17 is the most method-sensitive goal, sitting at rank 17 under LR but rank 8 under zero-shot; the LR baseline is the canonical reference, with MLP and zero-shot as uncertainty bounds (Appendix I.4).

Distance-functional robustness. The centroid gap is reproduced exactly as a validation gate; distribution-aware metrics (sliced Wasserstein, Chamfer, Gaussian-2-Wasserstein, MMD, energy, C2ST) recompute the gap on identical point clouds, holding assignment and embeddings fixed. Shape-sensitive metrics correlate only moderately with the adjusted centroid gap (e.g. $\rho = 0.525$ for sliced Wasserstein), and the Gaussian-2-Wasserstein shape term accounts for $\approx 79\%$ of the distance (Appendix C.2). The adjusted topic gap is therefore primarily a *mean-direction* (topic) finding; distributional shape carries a related but distinct signal, and after register removal that shape signal diverges from the centroid ranking.

Synthesis. Two conclusions survive the full cross-sensitivity grid. First, SDG 15 (Life on Land) is the robustly least-divergent goal: it sits at rank 17 (smallest gap) under the adjusted gap for LR and MLP across every encoder, and at rank 17 under the raw gap for most configs as well. Second, SDG 17 (Partnerships) is the most method-sensitive and the most divergent under the adjusted gap, while SDG 13 leads under the raw gap; the register decomposition explains the inversion (Section 4.2, Table 2). The central apparent cancellation (Section 4.3) replicates across every encoder–classifier config and policy-source family, so the findings are not an artefact of one measurement choice. Pooled OLS regressions that combine all 24 configurations into a single panel and control for encoder, retrieval, and segment-cap indicators confirm the adjusted-gap association with coverage divergence; full results are in Appendix J.

5 Discussion

The headline result is a possible cancellation: the raw coverage-vs-semantic-gap correlation is near-zero ($\rho = -0.012$, $p = 0.963$) because two opposing signals appear to cancel: topic divergence rises with coverage divergence ($\rho = 0.544$, $p = 0.024$) while register divergence falls with it ($\rho = -0.390$, $p = 0.122$), and the two would sum to near-zero if both components are real. The H1a test finds no detectable raw association in the canonical specification (Section 4; Table 3): with only $n = 17$ SDG-level observations the design is powered only to detect very large correlations ($|0.63|$ at 80% power), so the wide interval is consistent with either no effect or a moderate one and is evidence against a strong monotonic relationship, not proof of exact independence. All four hypotheses (H1a–H1d) are null on the raw gap across every encoder–classifier config (Table 3); after register removal the adjusted topic gap is positively associated with coverage divergence, so coverage and framing would be linked through topic if the register removal is valid.

This apparent cancellation is not a design confound. The supervised LR classifier (§3.5) is a labelling instrument with a transparent 768-coefficient boundary; the frozen Sentence-BERT space (§3.3) is a separate measurement surface. Because classification and measurement use different procedures, the apparent cancellation would describe the corpora's discourse structure. The encoder-sensitivity check confirms it: coverage ranks are nearly invariant across general-purpose and domain-specialised encoders ($\rho = 0.93$), while semantic-gap magnitude tracks whether the encoder separates research from policy registers.

The paper makes two contributions of unequal weight: a measurement framework separating classification from measurement and coverage from framing, which holds across changes of assignment method, policy source family, and segment cap (Section I.2, appendices); and a weaker set of AI–SDG findings, in which the coverage–semantic association reverses sign under some configurations (Table 3) and SDG 17's position depends heavily on assignment method (Table 14). The framework is the main contribution; the alignment findings are a first application to be checked against a larger, more diverse policy corpus, not settled results.

The coverage and gap profiles are conditional on corpus composition. The policy source-family split shifts the full profile: the curated AI/SDG sub-corpus is innovation-heavy (SDG 9 leads), while SDGi and UNGDC are climate-, partnerships-, and governance-oriented (SDGs 13, 17, 16) (Table 21, Appendix I.6). Claims are limited to findings that survive the cross-sensitivity filtering (Section 4.4).

5.1 Two Distinct Dimensions of Observed Divergence

If the cancellation holds, it is best read as a structural difference between knowledge systems, not a communication failure. The Mode 1/2 distinction (Gibbons et al., 1994) sharpens this: research concentrates on technically tractable applications in health, energy, and infrastructure (SDGs 3, 7, 9) where ML tasks fit well-defined problem-solving (Gohr et al., 2025; Vinuesa et al., 2020), while policy dominance of SDGs 13, 16, 17 reflects Mode 2 goal-setting from contexts of application demanding coordination. Epistemic-community theory bounds the claim (Haas, 1992): large within-SDG semantic gaps mark where shared problem-framing is less plausible, yet semantic proximity is only a possible precondition for knowledge transfer (Cross, 2013) and cannot show whether expert communities or policy uptake exist. The two-communities theory (Caplan, 1979) reinforces the reading: a near-zero raw correlation would be expected when the two corpora are distinct knowledge cultures with different reward systems.

5.2 Robust Patterns of Divergence

The cross-sensitivity grid (Section 4.4) narrows the gap findings to those stable enough for individual interpretation: SDG 15 is the robustly least-divergent goal and SDG 17 the most method-sensitive. SDG 17, the most divergent under the adjusted gap (0.326, Table 2), is consistent with its character as the means-of-implementation goal (Le Blanc, 2015). The register decomposition confirms that register uniformly inflates the raw gap, except for SDG 17 where register *increases* apparent similarity (-0.149).

SDG 3, SDG 16, and SDG 1 are the most robustly high-gap goals under the adjusted gap at the canonical MPNet–LR specification. The research corpus concentrates on technically tractable SDGs (3, 9, 4, 16, 11); SDG 4 is strongly represented but its share is partly inflated by ML learning-vocabulary (Appendix B.1). SDGs 7 and 12 sit consistently in the upper half of adjusted gap ranks.

Two examples make the gap concrete at the term level[2]. SDG 17 (adjusted gap 0.326, the widest after register removal), pairs technical (*model*, *method*, *machine*) with institutional (*countries*, *nations*, *global*, *agenda*) terms; its register component (-0.149) is the only negative value, indicating that register conventions make policy partnership discourse appear closer to research than the underlying topic warrants. SDG 13 (raw gap 0.480, the widest under the naive baseline), pairs research terms (*model*, *prediction*, *neural*, *analysis*) with policy terms (*climate*, *change*, *emissions*, *countries*, *action*): one measures, the other commits. The gap thus captures a real difference in how the two discourses frame shared SDG assignments, not a uniform property across goals.

5.3 Implications

Two implications argue against reading topical mention as alignment. **Agenda-setting**: the map does not prescribe funding redirection from embedding distances; it shows where research–policy translation and brokerage are most needed, and high-gap SDGs are candidates for boundary-organising practices that produce usable boundary objects accountable to both producers and users (Guston, 2001). **Research communication**: more output on neglected SDGs is insufficient without communication norms; funder-required policy-facing summaries, journal-level SDG justification fields, or policy synthesis briefs may raise semantic proximity more than topic-level mandates. Effective knowledge systems require simultaneous attention to salience, credibility, and legitimacy — a balance that topical mention alone does not achieve (Cash et al., 2003). For SDG monitoring bodies, funders, and journals alike, topical mention is not alignment: monitoring should report coverage and framing separately, allocation alone will not close the gap without policy-relevant communication, and SDG tags are not evidence of policy-aligned research.

Contribution back to scientometrics. The framework also contributes to the science-policy measurement literature itself. Existing SDG-mapping studies report coverage proportions — which SDGs are addressed and by how many papers — as the primary output. This paper demonstrates that coverage and framing are separable dimensions

[2]Distinctive terms are TF-IDF-weighted unigrams from deterministic samples assigned to each SDG by the LR classifier; full cases in Appendix C.1.

that can diverge independently (SDG 3 has the largest coverage gap but a near-median adjusted semantic gap; SDG 17 has the widest adjusted gap but moderate coverage). Scientometric studies that report only coverage risk conflating topical presence with policy-aligned framing (Armitage et al., 2020). Future SDG-mapping work should report both axes, following the cross-sensitivity structure of Table 13 as a template for bounding measurement uncertainty.

5.4 Limitations

Each scope condition below states the limit and the check that bounds it.

Semantic proximity, not substantive alignment or influence. The study measures embedding proximity, not whether research and policy are substantively aligned or whether research is used by policy. This is a scoping decision, not an unaddressed gap: the framework is positioned at Level 2 of a three-level construct and explicitly leaves Level 3 (substantive agreement) to qualitative follow-up.

Distance metric. Centroids discard distributional shape. Under raw embeddings the SDG ranking appeared preserved across the distribution-aware distances (Spearman $\rho = 0.44$–0.91), but this agreement was largely a register artifact: the distribution-aware metrics track the *register* component of the gap (which dominates the raw signal; variance ratio > 1), not the topic component (Appendix C.2). After register removal the adjusted topic gap is a mean-direction measure, and the distribution-aware metrics correlate only moderately with it (e.g. Gaussian-2-Wasserstein $\rho = 0.243$) while elevating SDGs 12 and 1 at the margin; they remain an informative full-distribution complement, not a re-ranking. The Wasserstein shape term accounts for $\approx 79\%$ of the distance. The only non-ranker is the classifier two-sample test (C2ST, AUC ≈ 1), a separation test (Appendix C.2). Sampling stability is verified with a replicate seed (maximum gap change 0.008).

Estimate stability under sampling. A 100-draw subsampling ladder (1k–2M papers) shows headline metrics stabilise well before the full corpus, so results are not sampling artefacts (Appendix D).

Corpus scope and retrieval. The policy corpus is international-institutional SDG discourse in English; the research query prioritises precision over recall (Section 3.1), minimising non-AI noise at the cost of specialised subfields. The source-family split makes composition sensitivity explicit (Appendix I.6). The comparison is inherently asymmetric: AI-for-sustainability research is a well-defined, if niche, academic category, whereas "AI and the SDGs" is not yet a mainstream organising category in policy discourse, so a policy corpus exactly comparable to the research intersection does not exist. I addressed this as best I could by re-testing the central interaction on the curated AI/SDG policy family, a symmetric AI/SDG-vs-AI/SDG construction (Section 4.3, Appendix I.6); broadening the research side to all-SDG research (dropping the AI-term intersection) would complete the symmetry and is left to future work. Between-SDG rankings are conditional on this asymmetric construction. The curated-AI/SDG policy family used for the symmetric check comprises only 94 documents, which is underpowered to confirm the asymmetric result with high confidence.

OpenAlex upstream classification. SDG tags inherit an upstream classifier's bias; untagged records are absent. This is a documented method-dependent boundary (Armitage et al., 2020), and coverage proportions are read conditional on it.

Assignment-method and encoder sensitivity. Gaps are conditional on assignment: the encoder- and assignment-method sensitivity tables show the supervised LR ranking is preserved across the two general-purpose encoders (MPNet vs. MiniLM LR $\rho \approx 0.63$), while the domain encoder SciBERT compresses gap magnitude to 0.04 yet preserves coverage structure ($\rho = 0.93$). The raw LR gaps are retained because they best match what the study measures and are most stable across source family and segment cap; MLP and zero-shot are the uncertainty bounds. The zero-shot method is scoped to one supervised-versus-nearest-centroid comparison under the canonical encoder (Appendix I.4); its near-zero correlation with the adjusted LR baseline ($\rho = 0.60$) is a method-axis effect (nearest-centroid assignment without a trained decision boundary), not an encoder one. The MLP robustness check agrees with zero-shot at the assignment level on the research corpus as well (Appendix I.4), so the two uncertainty bounds are mutually consistent.

Register effects. The gap may capture register, not topic. A register decomposition using Iterative Nullspace Projection (INLP) separates the raw centroid distance into a topic component (adjusted gap, mean 0.187) and a register component (mean 0.152; Appendix F). After register removal, the adjusted topic gap is the primary comparison (the INLP removal is evaluated against independent linguistic register markers in Appendix G): it shows a positive association with coverage divergence (Spearman $\rho = 0.544$, $p = 0.024$), while the raw gap is near-zero ($\rho = -0.012$, $p = 0.963$) because the register component ($\rho = -0.390$, $p = 0.122$) appears to cancel the topic signal. For all SDGs except SDG 17, the register component is positive (range: +0.155 to +0.261), confirming that register uniformly inflates the raw gap. The raw gap is retained as a reference measure because it directly corresponds to the observed framing difference, while the adjusted topic gap is the primary estimate. The removed subspace is interpreted as register; in principle, some substantive framing differences between corpora within an SDG could also be linearly decodable from corpus identity and removed alongside register. The independent-marker validation in Appendix G shows that the adjustment substantially reduces the corpus-level signal but does not eliminate it, and finds no robust evidence that topic is being systematically removed; the register interpretation therefore rests on partial, not complete, empirical support, and the validation's own limitations (register-score operationalisation, sample size, MPNet-only scope) are discussed there.

Structural precondition for register disentanglement. The INLP decomposition isolates register because the 17 SDGs supply a broad, near-orthogonal topic set that cross-cuts both registers: every goal has balanced academic and policy text, and the topics are distinct enough not to collapse onto a single direction. In domains where topic and register are tightly confounded (e.g., clinical notes versus biomedical literature, where shared terminology ties content to genre), the projection would leak topic into the register direction; the method's applicability is therefore bounded by the ontology, not the corpus.

Cross-sectional design. The study observes one period (2018–2025) against a similarly timed policy corpus; it cannot assess whether alignment is improving or deteriorating, or whether specific events (e.g., the EU AI Act 2024 (European Union, 2024), IPCC AR6 2022 (IPCC, 2022)) shifted research framing.

6 Conclusion

This paper introduces a measurement protocol that separates research-policy divergence into coverage and semantic framing, and demonstrates on the AI–SDG interface that the two are linked but separable. The primary result is the register-adjusted (INLP) topic gap, whose register interpretation is evaluated against independent linguistic register markers in Appendix G (limits in Section 5.4): under it SDG 17 is the most divergent goal, SDG 15 the least, and SDGs 3, 16, and 1 also strongly divergent; the raw gap inverts this ordering (SDG 13 leads raw), which is why the adjusted decomposition is the primary estimate. The raw coverage-vs-semantic-gap correlation is near-zero; the topic and register decompositions suggest this may reflect cancellation (the adjusted topic gap is positively associated with coverage divergence, while the register gap is negatively associated, though both component correlations are borderline at $n = 17$ SDGs). After INLP register removal, the adjusted topic gap is positively associated with coverage divergence, so topical attention does predict framing proximity once register is removed. Across three assignment methods, four policy source families, and two segment-cap configurations, the null raw result is robust and the register–topic decomposition replicates.

The protocol is portable to any research-policy interface that shares a category system and embeds both sides in a common semantic space, such as conservation science, health governance, or technology assessment. Its cross-sensitivity structure (Table 13) is a reusable template for bounding uncertainty: instead of reporting a single gap ranking, it makes the stability of every position transparent across the measurement choices analysts control.

The framework does not solve the alignment problem. It advances measurement from Level 1 (lexical correspondence) to Level 2 (observed semantic proximity) of a three-level construct (Section 2.2); semantic distance cannot show whether policy actors use research, whether epistemic communities exist, or whether divergence is productive or problematic. Level 3 remains the domain of qualitative, institutional, and deliberative methods. The AI-for-sustainability findings are a first application, not settled results.

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A Supplementary Methodology

A.1 Retrieval and Query Chain

The research corpus was retrieved from the OpenAlex API (Priem et al., 2022) using 68 queries formed by the cross-product of OpenAlex's native SDG work filter and four AI search terms (`artificial intelligence`, `machine learning`, `deep learning`, `neural network`), restricted to `publication_year > 2017`. The 2018 start date reflects the post-AlexNet (Krizhevsky et al., 2012) emergence of deep learning as the dominant paradigm and a research-uptake lag after the SDGs' adoption in 2015. OpenAlex SDG metadata defines the candidate universe; the SDG assignments used in the analysis are produced by the independently validated supervised classifier (Section 3.5). Complete query construction and credential handling are implemented in `1_code/0_fetch/fetch_openalex.py`.

The retrieval boundary is a deliberate scope condition. Elsevier's data-driven characterisation of AI research draws on over 700 field-specific keywords across some 600,000 documents (Hellwig et al., 2019), favouring recall. The OECD.AI Observatory methodology instead defines a paper as AI when it carries the Artificial Intelligence or Machine Learning field of study in the Microsoft Academic Graph (MAG), whose Fields of Study taxonomy OpenAlex later inherited and extended by mapping it to Wikidata identifiers (OECD, 2024; Priem et al., 2022; Wang et al., 2020). The present design prioritises precision: a four-term query keeps non-AI noise out of the embedding space at the cost of recall in specialised AI subfields. Schoeberl et al. (2023) reach the same conclusion for AI-research identification, namely that results are sensitive to the retrieval method and that supervised classification is preferred.

A direct test of this scope condition re-ran retrieval using field-of-study membership for the same two AI fields, mirroring the OECD.AI definition (OECD, 2024), and re-scored the resulting corpus through the identical pipeline. Concept retrieval used OpenAlex concept tags for Artificial Intelligence and Machine Learning, capped at `MAX_PAPERS_CONCEPT_CORPUS` = 100,000 papers, yielding the roughly 100,000-paper concept corpus. This scale sits on the empirical variance plateau established by the sample-stability ladder (Appendix D): at 50,000 papers the coverage-profile standard deviation is 0.001 and the mean within-SDG semantic gap is 0.342, within 0.002 of the full-corpus estimate of 0.340 (full corpus N = 3,105,144), while policy-text calibration bias has also converged (0.122). The scale also exceeds every corpus size used in prior SDG classification and mapping benchmarks, namely the OSDG community dataset (30,534 texts; Pukelis et al. (2022)), the SDG benchmark encoder evaluation corpus (35,001 papers; Gjorgjevikj et al. (2025)), the SDGi corpus (20,267 segments; SDGi (2024)), the Aurora expert-validated set (4,971 texts; Vanderfeesten et al. (2020)), and the 6,000-paper Kashnitsky et al. benchmark (Kashnitsky et al. (2024)), while remaining far smaller than the full 3,105,144-segment research corpus and thus a computationally tractable substrate for OpenAlex concept tagging.

A.2 Segmentation Mechanics

Each source document was segmented using the embedding model's own tokenizer: NLTK sentence-boundary detection (`sent_tokenize`) is followed by greedy accumulation of sentences up to a token budget of `max_seq_length - 10` (margin 10, empirically verified to reduce truncation to near zero). Segments falling below 20 words (`MIN_SEGMENT_WORDS`) are discarded; a single sentence exceeding the token budget is kept whole because mid-sentence splitting is not permissible. For the UN General Debate Corpus, paragraph-level keyword filtering precedes segmentation, so only SDG-relevant speech passages enter the pipeline; this pre-filtering is why UNGDC's segment count (6,128) is much smaller than its source-speech count. OSDG and the SDG Classification Benchmark bypass segmentation because they already consist of short single-label texts. The reference corpora (Knowledge Hub, SDGi-labels, Aurora) and the policy sources are segmented with the same procedure. Segment identifiers encode source document identity for document-level weighting; cross-source deduplication by exact text match (SDGi retained first) yields 40,597 unique policy segments.

A.3 Truncation Fix

During pipeline development, a truncation issue was discovered and corrected: texts exceeding the encoder's maximum sequence length were being silently truncated before embedding, discarding tail content. The fix uses the model's own tokenizer during segmentation (Appendix A.2) to enforce a token budget of `max_seq_length - 10`, verified by the `verify_truncation_rate` utility to produce near-zero post-segmentation truncation. The full corpus was re-embedded after the fix; all reported results use post-fix embeddings.

A.4 Reference-Corpus Provenance

The five reference corpora enter the pipeline at different sizes before preprocessing: OSDG 30,534 raw texts; SDG Classification Benchmark 616; IISD SDG Knowledge Hub 9,172; Aurora 4,022; SDGi 4,225 documents. After deduplication and English filtering, the pooled reference set feeds the single-label filter (Section 3.5), which retains only records with exactly one SDG label, yielding the per-source pool counts reported there (OSDG 30,532; SDGi 20,267; Knowledge Hub 6,122; Aurora 4,971; Benchmark 281) and the 62,173 total. OSDG and the Benchmark bypass segmentation, so their pool counts equal their deduplicated text counts; the other three corpora are segmented before pooling, which is why their pool counts are segment counts.

B Diagnostics for Reference Classifier

B.1 SDG 4 Lexical Artefact Audit

The SDG 4 concern is not used to relabel research records. Its narrower purpose is to test whether the concern is empirically plausible. Table 4 therefore compares the lexical composition of SDG 4-assigned research records with a matched non-SDG4 sample and with SDG 9, using only two transparent keyword groups: ML-learning vocabulary and education-specific vocabulary.

The audit points to a more nuanced conclusion than a pure false-positive story. More than half of the SDG 4-assigned set contains education-specific vocabulary without ML-only terms, and roughly another third contains both education and ML-learning vocabulary. By contrast, the matched non-SDG4 sample and the SDG 9 comparison set are dominated by ML-only vocabulary. This means the SDG 4 signal is not simply the same technical language that drives SDG 9. However, the large "both" share also confirms that ML-learning language and education language overlap materially within the SDG 4-assigned set.

The main analysis therefore retains the canonical LR argmax assignment but continues to treat SDG 4 as artefact-affected. The most accurate reading is a learning-related assignment whose interpretation is blurred by lexical overlap: neither "clean education" nor "pure ML artefact."

Table 4. SDG 4 lexical artefact audit.

Corpus subset	N	ML-only %	Education-only %	Both %	Neither / unclear %
SDG 4-assigned research	470002	11.8	44.4	23.8	19.9
Non-SDG4 research sample	470002	50.3	4.0	2.6	43.1
SDG 9-assigned research	577833	61.5	2.5	2.9	33.1

Notes: The audit is not used to reassign SDGs. It only checks whether SDG 4-assigned research is disproportionately dominated by machine-learning vocabulary, education vocabulary, or both.

B.2 Centroid Similarity

This subsection reports the pairwise cosine-similarity heatmap (Figure 7) for the 17 canonical SDG centroids. The centroids are the per-SDG mean training embeddings (L2-normalised) used by the LR classifier. The heatmap provides supplementary geometric context for the classification geometry that Section 3.5 builds on.

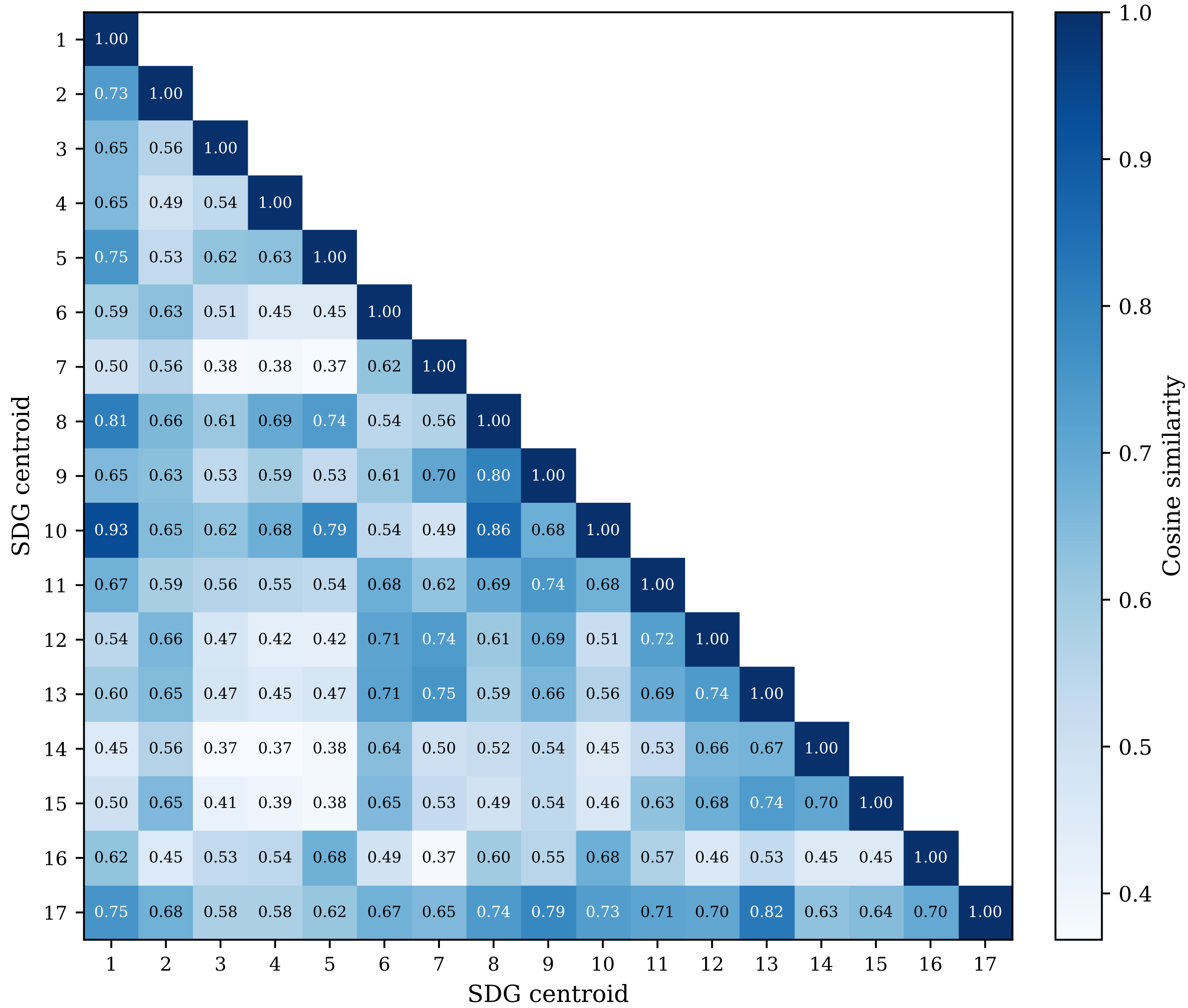

Figure 7. Pairwise cosine similarity between canonical SDG centroids (lower triangle). SDG 13 and SDG 17 sit close together (cosine 0.821), as do SDGs 1, 8, and 10.

Figure 7 shows the full 17×17 similarity grid. The lower triangle is annotated; the upper triangle is left blank because the matrix is symmetric. Reading the grid, three observations stand out. First, the diagonal dominates: most centroids are far more similar to themselves than to any neighbour, which is why the supervised LR classifier (Table 1) can build on this well-separated base to reach 0.816. Second, a small set of off-diagonal cells is elevated. SDG 11's centroid neighbours SDG 9 (cosine 0.744), and SDG 8 lies near SDGs 1 and 10 (cosine 0.808 and 0.862), and SDGs 1 and 10 are themselves the closest off-diagonal pair (cosine 0.931) – these are the proximities that make the SDG 11 and SDG 8 scores in Section 4.1 indicative rather than exact. Third, SDG 17 is the clear outlier: its row and column are the flattest, and its nearest neighbour is SDG 13 (cosine 0.821), reflecting the overlap between climate and partnership discourse noted in Section 4.1.

C Supplementary Robustness and Sensitivity Checks

C.1 Lexical Illustration of the Semantic Gap

Table 5 lists the TF-IDF terms most distinctive of the research and policy sides for each SDG. Generic ML methodology vocabulary, such as *model*, *learning*, and *neural*, is excluded because these terms reflect the search strategy rather than SDG-specific framing.

The contrasts echo the embedding patterns. The widest-gap SDGs (13, 11, 3, 1, 7, 8, 12; $\rho \geq 0.418$) consistently oppose technical measurement vocabulary, *detection*, *images*, *yield*, *regression*, against institutional or welfare terms, *climate*, *emissions*, *health*, *employment*. SDG 13 clarifies the pattern most sharply: research terms centre on crop monitoring and soil science while policy terms address national emissions commitments. Lower-gap SDGs (17, 15, 4, 6, 2, 9; $\rho \leq 0.292$) show narrower lexical divergence, such as the shared technology-infrastructure frame of SDG 9 or the social-media-partnership overlap of SDG 17. A further observation cuts across both groups: agricultural-AI vocabulary (*crop*, *soil*, *agricultural*) appears on the research side of nine unrelated SDGs, reflecting a strong agri-tech skew in the AI-for-SDG literature rather than alignment with each goal's policy domain.

C.2 Distance-Functional Robustness Table

Table 6 reports the per-SDG research–policy gap under the full battery of distribution-aware metrics described in Section C.2, together with the primary centroid gap. Part 1 covers the full-corpus exact metrics (sliced Wasserstein, Chamfer, Hausdorff, Gaussian-2-Wasserstein, Grassmann, linear MMD); part 2 covers the sampled metrics (exact EMD, Sinkhorn, energy, squared RBF MMD, C2ST AUC) plus the sampled centroid gap. The final row of each part reports Spearman ρ between each metric's ranking and the primary (adjusted) ranking.

Why the raw robustness result does not carry over. Under raw embeddings the distribution-aware metrics appeared to preserve the centroid ranking ($\rho = 0.44$–0.91). This agreement was driven by the shared *register* component, not by topic. Decomposing the raw gap into adjusted topic gap + register component, the register component has *larger* variance than the raw gap itself (variance ratio 1.24) and correlates $\rho = 0.67$ with the raw gap while correlating $\rho = -0.29$ with the adjusted topic gap. The distribution-aware metrics therefore tracked register, which is why they agreed with the raw centroid gap but diverge from the adjusted topic gap. After INLP removes register, the metrics instead track topic-space *shape*, which is why their correlation with the adjusted centroid gap falls to the 0.3–0.7 range and they elevate SDGs 12 and 1 at the margin. This is a measurement clarification, not a contradiction: the adjusted topic gap is a mean-direction estimate, and the distribution-aware distances are a complementary full-distribution view. Future work should report both and, where the two disagree, investigate whether the divergence is shape-driven (as here) or a residual artefact.

C.3 Implications for Interpreting the Main Estimates

These diagnostics are methodological stress tests that clarify the limits of the LR classifier approach, rather than failed repairs. The PCA diagnostics show that real-world SDG discourse is fuzzy rather than cleanly clustered, especially in the research corpus. The SDG 4 audit shows that the concern is real, but more nuanced than a pure machine-learning false positive. The policy source-family split shows that the full-corpus policy profile is a broader institutional SDG discourse rather than a universal AI policy profile. The canonical hard-threshold classifier assignment (argmax) is retained because it is transparent, reproducible, and directly interpretable.

These analyses also show that the LR classifier should be interpreted as a transparent semantic anchoring procedure rather than a definitive operational SDG system. Attempts to address multi-label overlap and corpus-register bias improve some aspects of the design but introduce new modelling choices. In this paper, the hard-threshold classifier assignment is therefore retained as the canonical specification, while the alternatives are reported as robustness and boundary-condition checks.

Table 5. Distinctive TF-IDF terms for all 17 SDGs.

SDG	Gap	Research-side distinctive terms	Policy-side distinctive terms
1	0.415	alleviation, china, approach, zakat, satellite, household, studies, indonesia, literature, relationship	social, cent, national, people, protection, services, population, vulnerable, living, million
2	0.299	crop, detection, plant, diseases, images, leaf, precision, techniques, soil, yield	population, food, children, prevalence, worst, access, hunger, million, births, ratio
3	0.494	patients, cancer, cells, images, clinical, detection, disease, potential, genes, predict	health, services, care, national, population, births, mortality, rate, people, access
4	0.267	cognitive, brain, neurons, sleep, agricultural, food, cells, motor, memory, poor	education, school, secondary, primary, national, quality, children, percent, access, vocational
5	0.278	menstrual, poor, pain, mothers, differences, relationship, patients, fertility, stress, sex	gender, violence, equality, national, rights, girls, men, cent, sexual, domestic
6	0.292	irrigation, soil, agricultural, crop, salinity, agriculture, sup, potential, sub, yield	sanitation, access, drinking, supply, national, wastewater, cent, services, population, clean
7	0.428	agricultural, poverty, agriculture, battery, optimization, approach, load, drying, poor, artificial	access, electricity, renewable, clean, worst best, population, affordable, index worst, cent, national
8	0.387	detection, image, food, recognition, leaf, plant, cnn, convolutional, disease, techniques	growth, employment, gdp, worst, labour, best, economic, cent, government, tourism
9	0.314	agricultural, agriculture, iot, detection, food, crop, plant, farming, farmers, soil	infrastructure, innovation, national, government, public, countries, strategy, access, population, intelligence
10	0.271	wealth, income inequality, effects, evidence, studies, que, distribution, literature, mobility, poor	people, rights, countries, migration, national, cent, disabilities, persons, discrimination, target
11	0.449	detection, image, accident, traffic, object, poor, recognition, spatial, fusion, agricultural	housing, city, urban, transport, cities, public, waste, national, plan, green
12	0.296	food, agricultural, detection, quality, potential, behavior, samples, organic, design, compounds	waste, consumption, national, recycling, production, procurement, management, public, economy, circular
13	0.480	soil, agricultural, crop, moisture, drought, sub, rice, stress, agriculture, yield	climate, change, emissions, countries, national, action, global, energy, nations, united
14	0.358	detection, images, aquaculture, food, bacterial, shrimp, cells, imaging, farming, techniques	marine, fish caught, biodiversity, rate, ocean, worst, population, sea, embodied, coastal
15	0.239	soil, crop, agricultural, plant, spectral, images, sensing, detection, remote, field	biodiversity, forest, worst, national, protected, population, area, freshwater, species, best
16	0.329	poor, food, covid-, pain, image, detection, social, studies, patients, brain	international, nations, peace, rights, united, security, human, law, world, global
17	0.177	sentiment, tweets, covid-, twitter, social, agricultural, dan, yang, zakat, media	countries, oecd, international, nations, united, agenda, report, financing, global, challenges

Notes: Terms are TF-IDF-weighted unigrams and bigrams from deterministic samples of research and policy texts assigned to each SDG.

Table 6. Per-SDG research–policy gap under distribution-aware metrics versus the canonical centroid gap (Part 1 of 2: full-corpus exact metrics). SWD = sliced Wasserstein; MMD^2 = squared RBF maximum mean discrepancy; Gauss. W_2 = Gaussian 2-Wasserstein; C2ST = classifier two-sample test AUC. The final row reports Spearman ρ between each metric's ranking and the canonical ranking across the 17 SDGs.

SDG	Canonical	SWD	Chamfer	Hausdorff	Gauss. W_2	Grassmann	linear MMD
1	0.191	0.009	0.509	0.600	0.684	3.727	0.067
2	0.089	0.008	0.388	0.454	0.618	3.629	0.057
3	0.319	0.011	0.454	0.581	0.717	3.717	0.129
4	0.158	0.010	0.399	0.487	0.607	3.307	0.096
5	0.151	0.010	0.429	0.529	0.646	3.514	0.095
6	0.130	0.009	0.417	0.475	0.639	3.598	0.080
7	0.170	0.010	0.427	0.522	0.687	3.729	0.105
8	0.184	0.008	0.474	0.564	0.658	3.566	0.065
9	0.228	0.010	0.430	0.556	0.671	3.444	0.104
10	0.195	0.008	0.456	0.517	0.617	3.343	0.062
11	0.209	0.009	0.446	0.542	0.677	3.484	0.091
12	0.156	0.009	0.467	0.566	0.677	3.462	0.094
13	0.219	0.010	0.417	0.498	0.667	3.667	0.117
14	0.132	0.010	0.415	0.510	0.675	3.753	0.098
15	0.084	0.008	0.394	0.464	0.578	3.421	0.056
16	0.242	0.010	0.417	0.520	0.640	3.333	0.103
17	0.326	0.009	0.461	0.490	0.595	2.993	0.104
ρ vs. canonical	–	0.525	0.485	0.444	0.243	-0.243	0.620

Table 7. Per-SDG research–policy gap under distribution-aware metrics versus the canonical centroid gap (Part 2 of 2: sampled metrics; continuation of Table 6). EMD = exact 1-Wasserstein; MMD^2 = squared RBF-MMD; Centroid (samp.) = sampled centroid-gap control column.

SDG	Canonical	EMD	Sinkhorn	Energy	MMD^2	C2ST AUC	Centroid (samp.)
1	0.191	0.645	0.623	0.056	0.025	0.969	0.186
2	0.089	0.533	0.516	0.054	0.026	0.979	0.088
3	0.319	0.665	0.597	0.105	0.046	0.985	0.315
4	0.158	0.554	0.523	0.084	0.040	0.988	0.157
5	0.151	0.581	0.543	0.082	0.038	0.984	0.152
6	0.130	0.546	0.547	0.073	0.036	0.986	0.128
7	0.170	0.581	0.545	0.094	0.045	0.990	0.170
8	0.184	0.620	0.596	0.054	0.024	0.978	0.185
9	0.228	0.625	0.560	0.090	0.040	0.984	0.228
10	0.195	0.602	0.636	0.051	0.023	0.964	0.191
11	0.209	0.606	0.580	0.076	0.035	0.977	0.213
12	0.156	0.605	0.605	0.081	0.037	0.987	0.154
13	0.219	0.577	0.485	0.102	0.048	0.983	0.217
14	0.132	0.555	0.538	0.091	0.044	0.984	0.134
15	0.084	0.515	0.515	0.053	0.026	0.982	0.083
16	0.242	0.604	0.495	0.088	0.039	0.983	0.241
17	0.326	0.582	0.499	0.083	0.037	0.980	0.319
ρ vs. canonical	–	0.694	0.047	0.444	0.260	-0.164	1.000

D Sample-Stability Robustness Check

To test whether the headline findings depend heavily on research-corpus size, I reran the downstream research-side analysis on repeated random subsamples drawn from the pre-embedded, pre-scored OpenAlex corpus. For each of 11 sampled tiers (from 1,000 to 2,000,000 papers), I drew 100 independent samples without replacement, recomputed the research coverage profile, rebuilt sampled research centroids, recalculated the mean within-SDG semantic gap, and re-estimated the policy-text calibration-bias summary. Policy-side quantities were held fixed. The full 3,105,144-segment analysis is shown as a deterministic anchor row rather than as another repeated sample.

Table 8. Sample-stability ladder for the research corpus.

Sample Size	Mean SD of SDG coverage shares	Mean within-SDG semantic gap	Policy-text calibration bias
1,000	0.006	0.417 ± 0.020	0.122 ± 0.008
2,000	0.004	0.384 ± 0.015	0.122 ± 0.006
5,000	0.003	0.358 ± 0.009	0.122 ± 0.004
10,000	0.002	0.350 ± 0.006	0.122 ± 0.002
20,000	0.001	0.344 ± 0.005	0.122 ± 0.002
50,000	0.001	0.342 ± 0.003	0.122 ± 0.001
100,000	0.001	0.341 ± 0.003	0.122 ± 0.001
200,000	0.000	0.340 ± 0.002	0.122 ± 0.001
500,000	0.000	0.340 ± 0.001	0.122 ± 0.000
1,000,000	0.000	0.340 ± 0.001	0.122 ± 0.000
2,000,000	0.000	0.340 ± 0.000	0.122 ± 0.000
Full corpus	–	0.340	0.122

Notes: Mean SD of SDG coverage shares is the unweighted mean of the per-SDG standard deviations in research coverage share across the 100 random draws at each sampled tier. The remaining columns report the mean and draw-to-draw standard deviation of the headline downstream metrics. The full-corpus row is deterministic and therefore has no variance term.

Plain-language guide.[3] The stability ladder shows that the clearest practical plateau appears by roughly 200,000 papers, where the mean semantic gap (∼0.340) and policy-text calibration bias (∼0.122) already stabilise to within 0.001 of their full-corpus values. From 500,000 papers upward, draw-to-draw dispersion is negligible. The full 3,105,144-segment result is retained as the primary estimate because it uses all available data while delivering the tightest values, but the substantive conclusions are not artefacts of sample size choice.

[3] The three metrics are: (1) mean SD of SDG coverage shares, how much the topic balance moves across repeated samples; (2) mean semantic gap, how different research and policy language are within the same SDG; (3) policy–research score gap, how much the scoring instrument favours policy text over research text regardless of content.

The larger corpus also enables per-SDG subgroup detail and tighter rank-order precision that the plateau minimum does not guarantee.

Balanced-subset rank stability. The ladder holds the policy side fixed while the research side grows, so within a tier the two sides are never size-matched. Because research (2,536,771 abstracts) outweighs policy (6,367 documents) by about 398:1, I also tested whether the within-SDG semantic-gap *ranking* survives when the research side is subsampled to a size comparable to the policy corpus. Across 100 independent draws at the 50,000-paper tier, the gap ranking reproduces the full-corpus ordering at Spearman $\rho = 0.986$ (SD 0.009), already $\rho = 0.936$ at 10,000 papers and $\rho = 0.990$ at 100,000 papers, converging to $\rho = 1.000$ at the full corpus. The research-side over-representation therefore does not determine which SDGs show large versus small semantic gaps.

E Model Selection: Grid Search Protocol and Rationale

E.1 Models Tested

Three model families were evaluated under 5-fold GroupKFold cross-validation (document-level grouping, not random splits) on the pooled training data (n=52,835). The test set (n=9,338) was never touched during any grid search.

- **Logistic Regression (LR)**: Interpretable per-dimension coefficients per SDG; fast; standard multiclass baseline.
- **MLP (PyTorch)**: Can capture non-linear interactions in the embedding space.
- **RF / XGB**: Considered but excluded because the embedding space is already a learned representation – bagged trees add ensemble complexity without a clear mechanism for improvement over LR/MLP, and would not produce interpretable coefficients per dimension.

E.2 LR Grid Search

The LR grid searched C $\in \{0.1, 1, 2, 3, 5, 10, 30, 100\}$ with L2 penalty and class_weight $\in$ $\{\text{None}, \text{balanced}\}$ (16 L2 configurations). The best L2 configuration was C=3.0, penalty=l2, class_weight=None (CV macro-F1 = 0.8101 ± 0.0020). C=1.0, C=2.0 and C=5.0 lie within ∼0.0004 of the optimum, while C=10.0 is ∼0.0026 below; C=30 and C=100 degrade clearly (0.804 and 0.797), confirming the optimum is a moderately regularised C≈3 and that weaker regularisation overfits. C=3.0 was selected as the empirical optimum. A separate L1 (lasso) spot-check (C=10) reached CV macro-F1 = 0.8066 ± 0.0069, below the L2 champion, confirming that L1 penalisation is not competitive for this embedding geometry (prior spot-check documented in 5_notes/MODEL_SELECTION.md).

E.3 MLP Grid Search

The MLP swept n_layers $\in \{2, 3, 4, 6\}$ $\times$ hidden_size $\in \{256, 384\}$ $\times$ lr $\in \{3\text{e}{-}4, 1\text{e}{-}3\}$ (16 configurations) with weight_decay=0 and dropout=0.3, trained with AdamW and early stopping. The best MLP (3 layers, 256 hidden, lr=3e-4) achieved CV macro-F1 = 0.8192 ± 0.0032. Depth 3 was optimal; depth 6 did not improve over depth 3, and depth 2/4 were within ∼0.001, so architecture is not the binding constraint. Learning rate had negligible impact: across the two learning rates the CV macro-F1 range at the best architecture was 0.816–0.819.

E.4 Unified Ranking and Final Selection

Rank	Config	CV macro-F1
1	MLP 3L/256h (best)	0.8192 ± 0.0032
2	MLP 4L/256h	0.8169 ± 0.0024
3–16	MLP (remaining 14 configurations)	0.815–0.819
17	LR C=3, L2	0.8101 ± 0.0020
18	LR C=1, L2	0.8087 ± 0.0025

Champion: LR (C=3.0, L2). The gap between LR and MLP is 0.0091 macro-F1, a modest difference insufficient to justify the complexity trade-off. MLP requires 3 hidden layers, AdamW, learning rate tuning, early stopping, and dropout selection (16 configurations swept). LR requires one free parameter (C), no hidden layers, no dropout, no random seed sensitivity. For this study's purpose – building an interpretable supervised reference to compare corpora, not a classifier built to label single documents – LR is the right choice. LR provides 768 signed coefficients per SDG class, each directly attributable to a specific semantic dimension; MLP's hidden layers diffuse this information across non-linear transformations. The macro-F1 numbers are reported honestly: LR 0.8101 ± 0.0020, MLP 0.8192 ± 0.0032, gap 0.0091 macro-F1. LR is chosen for transparency, not superior accuracy.

E.5 Held-Out Test Evaluation of the MLP Robustness Check

The MLP test macro-F1 (0.826) cited in Section 4.1 is a separate evaluation from the CV grid-search results above. After the CV sweep, the champion MLP architecture (3 layers, 256 hidden, lr=3e-4) was retrained on the full training pool (n=52,835) and evaluated once against the held-out test set (n=9,338), the same sanctioned single-use test evaluation applied to the LR champion. The resulting test macro-F1 of 0.826 is not directly comparable to the CV mean of 0.8192 ± 0.0032 from the grid search: the CV number is an average across five document-grouped folds on the training pool alone, while the test number is a single evaluation on a disjoint held-out sample after retraining on the full pool. Both numbers confirm that the LR–MLP gap is modest and that architecture choice is not the binding constraint on classification quality.

F Register Removal: Iterative Convergence and Cross-Config Replication

F.1 Iterative Nullspace Projection (INLP) Register Removal: Convergence

The full INLP procedure and identification argument are given in Methodology (Section 3.8). This appendix reports only the convergence behaviour. The adjusted results are the primary decomposition, not sensitivity diagnostics.

Convergence results. Held-out accuracy dropped from 0.978 (iteration 1) to 0.498 by iteration 62, at which point no further linear register separation was detectable. The Spearman rank correlation between the iteration 1 and iteration 62 gap vectors was 0.343: the additional removed directions shifted the relative SDG ordering substantially. The mean gap compresses from 0.340 to 0.187, a reduction of 40%, before plateauing. The Spearman ρ decays rapidly in the first 15 iterations (1.000 to 0.456) and then stabilises around 0.369, indicating that the first 15 orthogonal directions capture the bulk of the register signal.

The decomposition reveals a striking pattern for SDG 17 (Partnerships for the Goals). Under the raw centroid distance, SDG 17 shows the smallest gap (0.177, rank 17 of 17), suggesting near-identical framing. After register removal, the adjusted gap is 0.326 — the largest adjusted gap in the corpus (rank 1). The register component is -0.149, the only negative value, indicating that register actually *increases* the semantic similarity for this policy-dominant SDG: the policy discourse on partnerships uses register conventions that make it appear closer to research than the underlying topic content warrants. For all other SDGs, the register component is positive (range: +0.155 to +0.261), confirming that register uniformly inflates the raw gap. The headline correlation pattern is consistent with the apparent cancellation: the raw coverage-vs-gap correlation is near-zero ($\rho = -0.012$, $p = 0.963$) because the register component ($\rho = -0.390$, $p = 0.122$) appears to cancel the topic signal ($\rho = 0.544$, $p = 0.024$; Table 2).

F.2 Iteration-Level Diagnostic

Table 9 shows the per-iteration diagnostic: held-out classification accuracy, mean within-SDG gap, and rank correlation with the iteration 1 gap vector. The accuracy drops sharply in the first 15 iterations and then plateaus, confirming that the first 15 orthogonal directions capture the bulk of the register signal. The mean gap compresses and stabilises at a non-zero value, demonstrating that the semantic gap is not an artefact of corpus-level register differences.

Table 9. Stratified iterative register-removal diagnostic.

Iteration	Test acc.	Mean gap	Spearman ρ vs iter 1
1	0.978	0.315	1.000
2	0.938	0.304	0.988
3	0.895	0.290	0.985
4	0.864	0.282	0.983
5	0.831	0.279	0.976
10	0.747	0.238	0.721
15	0.685	0.206	0.456
20	0.593	0.195	0.397
30	0.558	0.188	0.400
40	0.535	0.186	0.360
50	0.523	0.187	0.343
62	0.498	0.187	0.343

Notes: Test acc. = held-out accuracy on a stratified 15% test split after projecting out the first k learned directions. Mean gap = average within-SDG semantic gap after k removals. Spearman ρ = rank correlation of the gap vector at iteration k versus iteration 1.

G Register Removal: Validation Against Independent Linguistic Register Markers

G.1 Purpose and validation question

The register adjustment of Section 3.8 removes, through Iterative Nullspace Projection, the subspace of the embedding space that is linearly decodable as corpus identity. The structural argument for reading that subspace as register is that every INLP iteration is trained on SDG-balanced research and policy embeddings: no linear direction can simultaneously encode SDG topic and corpus identity, so what remains linearly decodable is the corpus-level register difference. That argument guarantees *what* is removed — a corpus-separable linear component — but it does not by itself establish what that component is. This appendix evaluates the register interpretation empirically, using an independent surface-linguistic register score computed from the segment texts, following Biber-style register features (Biber, 1988).

The validation can establish three things: whether the adjustment removes a large corpus-separable component; whether the removal is accompanied by evidence that SDG (topic) structure is being destroyed; and whether apparent residual register signals survive controlled samples. It cannot establish that the removed subspace is identical to the measured linguistic register features. The register score used here is a small set of surface indicators, and the validation is single-encoder and sample-limited. The conclusion is correspondingly bounded: the evidence is consistent with a register reading and provides partial — not complete — empirical support for it.

G.2 Register operationalisation and sample construction

Sample construction. All results use the canonical MPNet track, 408 segments per sample (12 per SDG per corpus, seed 42), and the same frozen projection matrix G as the main analysis. Two sample constructions are used. The initial screen used per-SDG deduplication only (390 distinct source documents). An audit of that sample found that 7 source documents — global flagship reports (the SDSN Sustainable Development Reports 2024 and 2025, the UNDP Human Development Report 2021/22, the WHO report on ethics and governance of AI for health, the EU AI Act, and two UN SDG progress reports) — contribute 25 segments across 15 of the 17 SDGs. These segments are register outliers (mean sentence length 276.9 words vs. 35.6 for the remaining units; passive rate 3.3 vs. 9.8; first-person rate 19.0 vs. 5.9; nominalisation rate 20.0 vs. 38.8), and because segments of the same document sit close together in embedding space they can create spurious pooled correlations.

This discovery motivated a more controlled primary design rather than a correction of a flawed one. Every analysis below is reported on a one-per-parent rebuild (global deduplication by source document; 408 distinct parents, no multi-parent units), which caps each flagship document at a single segment. The rebuild is not a subset of the original sample: excluded flagship segments are replaced by other policy segments within the same SDG, drawn overwhelmingly from shorter national and thematic monitoring reports. The two samples therefore differ not only in clustering structure but partly in composition, and the mega-vs-non-mega feature contrast above documents that shift. I return to this composition change in the Limitations.

Register score. For every sampled segment I compute six register features per 1,000 words — hedge-word rate, deontic-modality rate, passive-voice frequency (a be-form immediately preceding a past participle, allowing one intervening adverb), mean sentence length, first-person-pronoun rate, and nominalisation rate — and take the first principal component of the standardised features as the register score. The orientation (positive loading on sentence length) is a sign convention only; the score's interpretation does not rest on any single feature, and sentence length in particular is composition-sensitive, as shown below.

Corpus feature contrasts. With the sample structure established, the feature differences can be read correctly. Policy segments are markedly more deontic (3.130 vs. 0.780 per 1,000 words) and research segments hedge more (1.427 vs. 0.854) and passivise more (10.303 vs. 8.453); these contrasts are stable across samples and point in the directions commonly associated with institutional-modality and academic writing register, respectively. First-person and nominalisation rates are roughly balanced at the corpus level (6.463 vs. 6.901; 39.003 vs. 36.367), the flagship segments being the outliers on both. Mean sentence length is the one feature whose corpus direction is not robust: policy segments appear much longer on the original sample (63.803 vs. 37.035 words), but this contrast is driven by the flagship-document segments, whose extraction-artefact-inflated mean (276.9 words) does not generalise, and it reverses direction once those segments are set aside. Sentence length was therefore treated as composition-sensitive and used to motivate the robustness checks below rather than as a load-bearing indicator of register.

G.3 Corpus discrimination before and after INLP

A 5-fold logistic-regression classifier distinguishes research from policy segments almost perfectly on raw embeddings (accuracy 0.909 on the original sample; 0.944 on the one-per-parent sample; Table 10). After register removal the accuracy falls to 0.505 on the original sample (95% CI [0.457, 0.553], binomial $p = 0.441$ against 0.5) and to 0.603 on the one-per-parent sample (95% CI [0.555, 0.649], $p = 1.9e - 05$). The adjustment therefore removes a substantial share of the corpus-linear signal. The removal is not complete: the primary one-per-parent value is significantly above chance.

The original-sample value of 0.505 is not distinguishable from chance, but this holds only for the mega-document-contaminated composition. Re-estimating on the original sample with the 25 mega-document policy segments excluded raises adjusted accuracy to 0.574 (95% CI [0.524, 0.623], $p = 0.002$), and a bootstrap comparison of the two samples (difference +0.098, 95% CI [0.025, 0.169]) indicates that most of the rise is attributable to mega-document clustering: excluding the clustered policy segments closes most of the gap between the original and the one-per-parent samples. Residual above-chance separation after removal is not a failure of the method. INLP's stopping rule operates on held-out accuracy within its own training scheme, which converges on the training distribution; the validation classifier here is re-estimated from scratch on fresh sample draws, so the two evaluations are not directly comparable, and nothing in the method guarantees the validation value to be at chance.

The register-only rows of Table 10 bound the interpretation. A classifier trained on the six surface features alone reaches only 0.456 on the original sample and 0.544 on the one-per-parent sample — barely above chance in the latter case. The linguistic markers therefore capture only a part of the corpus distinction. This is consistent with the register reading (the corpora do differ in register-like surface ways), but it also delimits it: the removed embedding component is far broader than these six surface indicators, so the validation cannot claim that the removed subspace is exhausted by, or identical to, the measured features.

G.4 Topic preservation

If the adjustment removed topic, 17-way SDG selectivity — how well the SDG of a segment can be predicted from its embedding — would fall. It does not (Table 11): on the original per-SDG-dedup sample (draw 1), LR cross-validated accuracy is 0.691 on raw and 0.672 on adjusted embeddings, and kNN 0.554 and 0.578 (chance 0.059). The change is small and of mixed sign, and is best read as sampling noise around an unchanged quantity rather than as evidence of preservation in the strong sense. The correct statement is that there is no evidence that topic structure is systematically removed; the analysis does not claim that topic is preserved perfectly, and it was run on the original sample only, not re-estimated on the one-per-parent rebuild. Topic remains linearly decodable after register removal, as the identification argument anticipates.

G.5 Residual register diagnostics and robustness investigation

This section reports the robustness investigation as it unfolded, because its two outcomes cut in opposite directions and both are informative.

Two initial diagnostics on the original sample appeared to show residual register structure after adjustment. First, the register score correlated positively with the amount of embedding removed (Spearman $\rho = +0.102$, $p = 0.040$; within-research $\rho = +0.212$, within-policy $\rho = +0.191$). Second, distance to the SDG centroid in adjusted space correlated more strongly with the register score than in raw space ($\rho = +0.126$ rising to $\rho = +0.247$), which would suggest register remained partially decodable after removal. On the clean one-per-parent sample both signals disappear or reverse: the removed-norm correlation becomes $\rho = +0.092$ (n.s.; within-corpus -0.043 and -0.036, both n.s.), and the centroid-distance correlation becomes negative (raw $\rho = -0.212$; adjusted $\rho = -0.197$). The original patterns are therefore attributable to the flagship-document clustering — register-extreme, mutually similar segments of a few documents dominating the pooled correlation — not to residual register.

It is important to acknowledge both consequences of this collapse. It removes the apparent red flag of residual register surviving adjustment, but it also removes the positive evidence that the amount removed tracks the register score: the association that linked removed magnitude to the surface-linguistic markers on the original sample does not hold on the controlled sample. The direct surface-linguistic corroboration of the removed subspace is therefore weaker than the initial screen suggested, and the register reading rests on the corpus-discrimination result and on the absence of counter-evidence rather than on a measured correspondence between the removed component and the six features.

A within-SDG decomposition (register score vs. distance to the *other-corpus* SDG centroid, adjusted space) found one apparently surviving trace among policy segments ($\rho = -0.197$, $p = 0.005$). This value reproduced exactly on an independent re-analysis of the same sample — a re-run, not a fresh random draw — and, for that draw, survived mega-document exclusion ($\rho = -0.213$, $p = 0.003$). Before treating it as evidence, the one-per-parent construction was re-sampled with fresh random streams (seeds 43–45) as a draw-stability check. The statistic is draw-unstable: $\rho = -0.130$ ($p = 0.063$), $\rho = -0.004$ ($p = 0.949$), and $\rho = +0.126$ ($p = 0.072$), with the sign flipping across draws and the mean near zero. Mega-document exclusion is likewise draw-specific: the value after exclusion is $\rho = +0.127$ (n.s.), $\rho = -0.138$ (n.s.), and $\rho = -0.213$ ($p = 0.003$) on draws 1–3 respectively. Per-SDG, only 0 of 17 SDGs reach $p < 0.05$ (14 negative, 3 positive). The apparent trace is therefore treated as draw-level noise, not as residual register structure. This history is reported deliberately: each apparent signal in this validation was followed until it could be attributed either to flagship-document clustering or to draw instability, and none survived the controlled samples.

G.6 Validation conclusion

The validation supports the following claims. (i) The adjustment removes a substantial corpus-level component: corpus accuracy falls from 0.909 to 0.505 on the original sample and from 0.944 to 0.603 on the one-per-parent sample. (ii) The removed component is not strongly attributable to SDG topic loss: 17-way SDG selectivity is essentially unchanged. (iii) Apparent residual register signals were explained by sampling and document-clustering effects rather than by stable residual structure: the initial removed-magnitude and centroid-distance associations vanished or reversed on the controlled sample, and the one surviving within-SDG trace was draw-unstable. (iv) The remaining evidence provides partial empirical support for interpreting the removed component as register-related: the corpora differ in register-like surface features in stable directions, and the adjustment removes corpus-separable signal while leaving topic structure intact, but the direct correspondence between the removed component and the measured linguistic markers was not established, and residual corpus-linear signal remains after removal.

The register interpretation is therefore consistent with the validation evidence rather than confirmed by it. The conclusion should be read alongside the main text, which describes the register interpretation as resting on partial, not complete, empirical support.

G.7 Limitations

The validation is bounded in ways that should be read alongside the main text. First, the register score is operationalised as the first principal component of six hand-specified features, and the choice of operationalisation is not resolved here: an alternative a-priori "institutional" z-score combination gave null or reversed results on the initial screen, and the score is oriented by a sign convention rather than by a substantive prior. Second, each sample contains only 408 segments (12 per SDG per corpus), and the per-SDG tests within the draw-stability stage have $n = 12$ policy units each, so the 0-of-17 significant per-SDG result is not evidence of absence at the per-SDG level. Third, the draw-stability check was applied to the within-SDG trace only; the corpus-level accuracies in Table 10 were not re-estimated on fresh draws. Fourth, the one-per-parent sample is a rebuild, not a subset of the original sample, and the excluded flagship documents are replaced by shorter national and thematic monitoring documents, so the two-sample comparison is partly compositional; the mega-vs-non-mega feature contrast documents that shift, and the selectivity check was run on the original sample only. Fifth, the validation is MPNet-canonical only. Sixth, the residual corpus-linear signal that survives adjustment is not characterised further here; its exact linguistic content remains open, which is part of why the empirical support is partial rather than complete. Finally, two distinct statistics in this appendix round to -0.197 at three decimals — the one-per-parent pooled centroid-distance correlation (-0.197) and the draw-stability-stage policy other-corpus distance correlation (-0.197); they are different quantities and should not be conflated.

Table 10. Corpus discrimination accuracy (research vs. policy, 5-fold logistic regression) before and after INLP register removal.

Sample	Input	n	Fold-mean acc.	Pooled acc. (k/n)	Wilson 95% CI	Binom. p
Original (per-SDG dedup)	register-only	408	0.456	0.456 (186/408)	[0.408, 0.504]	0.967
Original (per-SDG dedup)	raw embeddings	408	0.909	0.909 (371/408)	[0.878, 0.933]	< 0.001
Original (per-SDG dedup)	adjusted (INLP)	408	0.505	0.505 (206/408)	[0.457, 0.553]	0.441
One-per-parent	register-only	408	0.544	0.544 (222/408)	[0.496, 0.592]	0.042
One-per-parent	raw embeddings	408	0.944	0.944 (385/408)	[0.917, 0.962]	< 0.001
One-per-parent	adjusted (INLP)	408	0.603	0.603 (246/408)	[0.555, 0.649]	< 0.001
Original (per-SDG dedup)	adjusted, mega-policy excluded	383	0.574	0.574 (220/383)	[0.524, 0.623]	0.002

Notes: Pooled accuracy = proportion of correct predictions across the five test folds. Wilson 95% CI and one-sided binomial test against 0.5 on the pooled proportion. In the one-per-parent sample each segment comes from a distinct source document, so no test segment shares a source document with its training folds; in the original per-SDG-dedup sample, segments of the same flagship document can straddle folds, so the original-sample rows carry a mild grouped-similarity caveat. Mega-policy excluded = original sample minus the 25 policy segments contributed by the 7 multi-segment documents.

Table 11. 17-way SDG (topic) selectivity before and after register removal (original per-SDG-dedup sample, draw 1).

Input	LR CV acc.	kNN(5) CV acc.	Chance
Raw	0.691	0.554	0.059
Adjusted (INLP)	0.672	0.578	0.059

Notes: Five-fold stratified cross-validated accuracy of logistic regression ($C = 10$) and kNN ($k = 5$) in predicting the SDG of each segment from raw and adjusted embeddings; chance = 1/17. If the adjustment removed topic, these accuracies would fall.

H Concept-Retrieval Sensitivity

Advisor review flagged the four-term keyword query used to retrieve the AI research corpus as an intuitive choice. To test sensitivity to that choice, retrieval was repeated using field-of-study membership for the two AI fields of study (OECD, 2024), mirroring the OECD.AI Observatory methodology, which treats a paper as AI if tagged with the Artificial Intelligence or Machine Learning field of study in the Microsoft Academic Graph (MAG), the taxonomy that OpenAlex later inherited and extended (Priem et al., 2022; Wang et al., 2020). No SDG filter was applied. SDG assignment was performed by the same trained logistic-regression classifier used throughout (Section 3.5), so the only difference from the main corpus is retrieval strategy. The resulting 100,000-paper corpus was passed through the identical segmentation, embedding, scoring, and analysis pipeline.

Table 12 compares the SDG coverage shares of the keyword-retrieved and concept-retrieved corpora. The two profiles are closely aligned (Spearman $\rho = 0.948$, Kendall $\tau = 0.868$); the largest deviations are SDG 4 (-7.6pp, the known ML-vocabulary artefact, Appendix B.1) and SDG 9 ($+6.4$pp). The within-SDG coverage-gap and semantic-gap rankings are also stable against the keyword baseline (coverage-gap rank $\rho = 0.87$; semantic-gap rank $\rho = 0.81$, Table 13). Schoeberl et al. (2023) independently find that AI-research identification is sensitive to the retrieval method and recommend supervised classification. The present LR scoring is that recommended step applied to SDG assignment. We therefore report the main findings as robust to the query-term choice, while noting that both corpora remain conditional on their respective retrieval strategies.

Table 12. SDG coverage-share comparison: keyword-retrieved versus concept-retrieved (AI/ML field-of-study) research corpus.

SDG	Keyword %	Concept %	Δ
SDG 1	0.4	0.7	+0.3
SDG 2	2.4	1.8	-0.6
SDG 3	25.7	25.7	-0.0
SDG 4	15.6	8.3	-7.3
SDG 5	2.0	1.7	-0.2
SDG 6	1.0	0.8	-0.2
SDG 7	4.1	4.7	+0.6
SDG 8	2.2	2.3	+0.1
SDG 9	20.1	25.9	+5.8
SDG 10	0.9	0.8	-0.1
SDG 11	5.1	8.0	+2.9
SDG 12	1.5	2.7	+1.2
SDG 13	3.7	4.2	+0.5
SDG 14	1.6	1.3	-0.3
SDG 15	2.2	2.1	-0.1
SDG 16	11.2	8.8	-2.4
SDG 17	0.3	0.2	-0.0
Spearman $\rho = 0.948$; Kendall $\tau = 0.868$			

Notes: Δ is concept minus keyword share in percentage points.

I Supplementary Cross-Method Data

I.1 Cross-Method Coverage and Semantic Gap Values

Tables 17 and 18 report the raw coverage gap and semantic gap values for each SDG across seven encoder–assignment-method combinations, with an additional concept-retrieval LR column for comparison against the keyword-retrieved research corpus baseline (Appendix H). While Tables 13 and 16 rank these values to compare cross-method stability, the tables below show the unrounded gap magnitudes, allowing the rank separation to be interpreted in absolute terms. Coverage gap is document-weighted (Assumption A19); semantic gap policy centroids use a 50-segment-per-document cap. The zero-shot nearest-centroid method is reported for the canonical MPNet encoder only; it is scoped to a single supervised-versus-nearest-centroid comparison (Appendix I.4), not a cross-encoder axis.

I.2 Rank Robustness: Cross-Sensitivity and Encoder Sensitivity

Table 13 reports each SDG's semantic-gap rank under the canonical MPNet–LR configuration and six alternatives spanning policy source family (curated AI/SDG, SDGi, UNGDC), segment cap (20, none), and retrieval strategy (concept-based field-of-study). Table 14 and Table 15 report the within-SDG gap rankings under MPNet, MiniLM, and SciBERT for each assignment method. In every rank-robustness table the adjusted (register-removed) ranking is Panel (a) and the raw (naive) ranking is Panel (b), with the tested dimension nested inside each panel.

I.3 Encoder Sensitivity

The semantic-gap ranking is moderately preserved across the three encoders (MPNet–LR vs. SciBERT–LR Spearman $\rho = 0.54$; MPNet–LR vs. MiniLM–LR $\rho \approx 0.63$), and the coverage-gap ranking is highly stable (MPNet–LR vs. SciBERT–LR $\rho = 0.93$). MiniLM and SciBERT are scored on a deterministic 100,000-paper subset (seed 42) of the canonical segmented corpus, so the architecture comparison isolates encoder choice from corpus scale. SDG 13 leads and SDG 17 trails under both general-purpose encoders; the scientific-domain SciBERT compresses the magnitudes and, at its LR assignment, shifts the lead to SDG 3 and the trail to SDG 8. Assignment-method sensitivity within MPNet: the three assignment methods, LR, zero-shot nearest-centroid, and the MLP robustness check, show varied agreement (MPNet LR vs. MLP $\rho = 0.85$, LR vs. zero-shot $\rho = 0.60$). SDG 17 is the most method-sensitive goal, sitting at rank 17 under LR but rank 8 under zero-shot. The LR baseline is the canonical reference, with MLP and zero-shot as uncertainty bounds.

I.4 Assignment-Method Comparison: Supervised vs. Nearest-Centroid

The zero-shot nearest-centroid method is used in this study for one purpose only: a single comparison between the canonical supervised assignment (LR, with the MLP as robustness) and an assignment rule with no trained decision boundary. Under the zero-shot rule, every research paper and policy segment is assigned to the SDG whose train-split reference centroid is most similar (cosine) in embedding space. Table 19 reports how often the two methods agree, per SDG and overall, for the research corpus (paper level) and the policy corpus (segment level), together with the LR-versus-zero-shot semantic-gap rank delta per SDG.

The comparison confirms that the two methods agree on the majority of assignments: the overall LR-versus-zero-shot agreement rate is 62.3% over the 3.1M research papers and 79.7% over policy segments (80.4% at document level; Table 19). Agreement is not uniform across SDGs, and the disagreements are concentrated where the reference centroids are least separated, in line with the centroid-similarity diagnostics (Appendix B.2). SDG 17, whose reference centroid is the flattest and most isolated in that analysis, is also the most method-sensitive goal here: it shows the largest rank delta between the two semantic-gap rankings ($|\Delta| = 9$) and one of the lowest per-SDG assignment agreement rates on the research side (16.8%, second only to SDG 10 at 15.9%). SDG 8, whose supervised rank is also heavily method-dependent (rank 6 under LR, rank 14 under zero-shot), shows the second-largest rank delta ($|\Delta| = 8$). The zero-shot rule, having no supervised decision boundary, anchors partnership and inequality discourse less sharply than the LR classifier. The MLP robustness check agrees with the zero-shot rule at 76.8% of policy segments (78.1% at document level), close to the LR-versus-zero-shot rate; research-level MLP agreement is not reported because MLP per-paper scores are not persisted, and the research LR-versus-MLP gap-rank correlation (Section I.3) proxies it.

I.5 Raw-Value Correlation: Metric-Sensitivity Companion

Table 20 repeats the H1a–H1d grid of Table 3 with one change: cells report Pearson r on the raw coverage and gap magnitudes instead of Spearman ρ on their ranks; the grid, predictors, and columns are identical. The four H1 hypotheses are co-equal and each probes a distinct coverage-structure prediction. Under both metrics the adjusted-gap association is stable across the supervised tracks for H1a and H1d (the hypotheses with a detectable signal: positive in 7/9 and 9/9 configs respectively under Pearson) and remains null for H1b and H1c (no consistent signal under either metric – itself informative: coverage divergence tracks the gap and policy share, not dominance or raw research share). The only reversals are the two SciBERT configs; the zero-shot config, already scoped as an uncertainty bound, is stable on the adjusted gap. SciBERT is designated a lower-trust robustness encoder (its scientific pretraining compresses gap magnitudes, 0.04 against 0.34, Section I.3).

Possible causes of the SciBERT divergence. (A) At $n = 17$ the minimal detectable Pearson r is large ($|\rho| \geq 0.63$ at 80% power); both SciBERT adjusted-gap values sit below it and are indistinguishable from zero, so the rank/magnitude sign flip is within estimation error. (B) Pearson weights absolute distances, Spearman weights ranks; SciBERT's $\sim 8\times$ smaller gap magnitudes make Pearson disproportionately sensitive to its few larger gaps. (C) SciBERT has the lowest assignment fidelity of the three encoders, giving noisier per-SDG gaps. (D) The flip is in the *adjusted* gap, and INLP removal is already documented as encoder-dependent. (E) Pearson is leveraged by individual outliers (e.g. SDG 3). None of A–E is established; A and E are the most directly testable. All are consistent with the divergence being uninformative for the conclusion, which rests on the supervised tracks where the signal is large (+0.70) and stable across metrics. The raw-value specification therefore confirms, rather than threatens, the rank-based result.

I.6 Detailed Policy Profiles by Source Family

The full policy corpus combines a curated AI/SDG sub-corpus, SDGi VNR/VLR reports, and UN General Debate speeches. These are all policy-facing texts, but they are not interchangeable genres. The family split reported in Table 21 therefore asks a narrow question: whether the full-corpus policy profile is broadly stable across source families or whether it is strongly shaped by corpus composition.

The answer is mixed. The broad international-institutional profile is clearly source-family sensitive. The curated AI/SDG family is much more innovation-oriented, with SDG 9 leading its document-weighted profile, while the SDGi and UNGDC families are more strongly concentrated on climate, partnerships, governance, and, in the SDGi case, SDG 11-style institutional review language. At the same time, the broader full-corpus pattern is not driven by one family alone: both the UNGDC speeches and the full corpus are dominated by the climate–partnerships–governance block, and SDG 17 remains the largest semantic-gap case across all three families.

The implication is interpretive rather than corrective. The full policy corpus should be read as international institutional SDG discourse, not as a universal profile of all AI sustainability policy. The scope of what the policy-side distribution represents is accordingly narrower, but the main diagnostic purpose remains valid.

The symmetric construction preserves the cancellation at the level of the interaction test. The H1a–H1d tests ask whether SDG coverage measures predict the within-SDG semantic gap (Section 4.3). Re-running those tests within each policy source family (Table 22) reproduces the full-corpus null under the curated construction: with policy restricted to the curated AI/SDG family, research share does not correlate with the gap ($\rho = -0.06$, $p = 0.830$), and the coverage-gap correlation also does not reach significance ($\rho = -0.22$, $p = 0.390$). The full-corpus row of Table 22 reproduces the canonical interaction estimates (Section 4.3), confirming the replication is faithful. The null is therefore not an artefact of comparing an AI-specific research intersection with a broader policy union.

Table 13. Cross-sensitivity robustness of within-SDG semantic-gap rankings across policy source, segment cap, and retrieval strategy.

SDG	Canon	Segment cap 20	Segment cap None	Retrieval LR	Retrieval MLP
SDG 1	8	8	8	4	6
SDG 2	16	16	17	17	17
SDG 3	2	1	2	3	5
SDG 4	11	11	11	14	14
SDG 5	13	13	13	6	10
SDG 6	15	15	15	15	16
SDG 7	10	10	9	13	13
SDG 8	9	9	7	5	1
SDG 9	4	5	3	7	7
SDG 10	7	7	10	9	3
SDG 11	6	6	5	2	2
SDG 12	12	12	12	10	11
SDG 13	5	4	4	11	12
SDG 14	14	14	14	12	9
SDG 15	17	17	16	16	15
SDG 16	3	3	6	8	8
SDG 17	1	2	1	1	4
Rank Corr (ρ)	1.00	1.00	0.97	0.75	0.68

Panel (a): Adjusted (register-removed, canonical).

SDG	Canon	Policy source				Segment cap		Retrieval	
		Curated	SDGi	UNGDC	Full	20	None	LR	MLP
SDG 1	5	2	5	5	5	5	5	1	1
SDG 2	10	9	11	11	10	10	9	11	10
SDG 3	1	8	1	1	1	1	2	3	7
SDG 4	15	15	12	15	14	15	14	13	17
SDG 5	13	13	14	13	13	13	13	6	5
SDG 6	12	10	13	12	12	12	12	15	15
SDG 7	4	1	6	2	4	4	4	8	11
SDG 8	6	5	7	6	6	6	6	4	2
SDG 9	9	14	2	10	9	8	8	12	13
SDG 10	14	11	15	16	15	14	15	10	8
SDG 11	3	4	3	4	3	3	3	2	3
SDG 12	11	6	10	8	11	11	11	14	12
SDG 13	2	3	4	3	2	2	1	5	4
SDG 14	7	7	9	9	7	7	7	9	6
SDG 15	16	12	16	14	16	16	16	16	16

Table 14. Domain-encoder sensitivity of within-SDG semantic-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

SDG	Encoder (embedding architecture)						
	MPNet (768-d)			MiniLM (384-d)		SciBERT (768-d)	
	LR	MLP	ZS	LR	MLP	LR	MLP
SDG 1	**8**	**11**	**1**	**8**	**7**	**15**	**1**
SDG 2	**16**	**16**	**14**	**17**	**17**	**17**	**16**
SDG 3	**2**	**1**	**3**	**2**	**1**	**5**	**12**
SDG 4	*11*	*14*	*13*	*7*	*8*	*9*	*11*
SDG 5	*13*	*10*	*11*	*1*	*3*	*2*	*6*
SDG 6	**15**	**15**	**10**	**16**	**15**	**14**	**17**
SDG 7	10	12	8	12	12	7	10
SDG 8	9	4	12	11	11	13	14
SDG 9	*4*	*9*	*16*	*10*	*9*	*11*	*15*
SDG 10	7	5	4	4	4	6	2
SDG 11	*6*	*6*	*9*	*13*	*13*	*4*	*5*
SDG 12	12	7	5	9	10	16	8
SDG 13	**5**	**8**	**6**	**6**	**5**	**3**	**4**
SDG 14	**14**	**13**	**15**	**15**	**14**	**10**	**7**
SDG 15	17	17	17	14	16	12	13
SDG 16	**3**	**3**	**7**	**3**	**2**	**1**	**3**
SDG 17	*1*	*2*	*2*	*5*	*6*	*8*	*9*
Rank Corr (ρ)	1.00	0.85	0.60	0.63	0.71	0.54	0.36

Panel (a): Adjusted (register-removed, canonical).

SDG	Encoder (embedding architecture)						
	MPNet (768-d)			MiniLM (384-d)		SciBERT (768-d)	
	LR	MLP	ZS	LR	MLP	LR	MLP
SDG 1	*5*	*8*	*7*	*17*	*17*	*16*	*1*
SDG 2	*10*	*11*	*10*	*14*	*14*	*14*	*14*
SDG 3	**1**	**4**	**4**	**2**	**2**	**1**	**3**
SDG 4	**15**	**15**	**15**	**15**	**15**	**10**	**12**
SDG 5	*13*	*9*	*16*	*4*	*8*	*11*	*15*
SDG 6	**12**	**14**	**5**	**13**	**16**	**6**	**8**
SDG 7	**4**	**7**	**1**	**5**	**4**	**4**	**4**
SDG 8	*6*	*3*	*14*	*10*	*12*	*15*	*11*
SDG 9	**9**	**12**	**17**	**8**	**6**	**7**	**13**
SDG 10	*14*	*13*	*12*	*3*	*3*	*5*	*7*
SDG 11	3	2	2	6	7	3	2
SDG 12	*11*	*5*	*9*	*7*	*5*	*13*	*5*
SDG 13	**2**	**1**	**6**	**1**	**1**	**2**	**6**
SDG 14	*7*	*6*	*3*	*11*	*10*	*8*	*9*
SDG 15	*16*	*17*	*11*	*9*	*13*	*9*	*10*
SDG 16	*8*	*10*	*13*	*12*	*11*	*12*	*16*
SDG 17	**17**	**16**	**8**	**16**	**9**	**17**	**17**

Table 15. Domain-encoder sensitivity of within-SDG coverage-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

SDG	Encoder (embedding architecture)						
	MPNet (768-d)			MiniLM (384-d)		SciBERT (768-d)	
	LR	MLP	ZS	LR	MLP	LR	MLP
SDG 1	**7**	**6**	**7**	**7**	**7**	**7**	**6**
SDG 2	*15*	*13*	*14*	*11*	*11*	*11*	*9*
SDG 3	**1**	**1**	**2**	**1**	**1**	**1**	**1**
SDG 4	**4**	**4**	**3**	**5**	**4**	**3**	**3**
SDG 5	*8*	*10*	*8*	*12*	*9*	*8*	*8*
SDG 6	**9**	**8**	**12**	**8**	**10**	**12**	**10**
SDG 7	**16**	**17**	**6**	**16**	**17**	**17**	**17**
SDG 8	11	15	9	9	8	9	11
SDG 9	**2**	**2**	4	**2**	**2**	**2**	**4**
SDG 10	**14**	**9**	**11**	**15**	**14**	**15**	**14**
SDG 11	*17*	*16*	*15*	*13*	*13*	*14*	*13*
SDG 12	12	14	13	14	15	13	15
SDG 13	**5**	**5**	**5**	**6**	**6**	**6**	**5**
SDG 14	**10**	**12**	**16**	**10**	**12**	**10**	**12**
SDG 15	*13*	*11*	*17*	*17*	*16*	*16*	*16*
SDG 16	6	7	10	3	3	5	7
SDG 17	**3**	**3**	**1**	**4**	**5**	**4**	**2**
Rank Corr (ρ)	1.00	0.92	0.75	0.89	0.90	0.93	0.90

Notes: Each cell reports the within-SDG coverage-gap rank (1 = largest gap, 17 = smallest gap) under that encoder and assignment method. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section I.3). LR = canonical supervised logistic regression; MLP = 4-layer/384-hidden network; Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only. Coverage gap is document-weighted (Assumption A19) for all methods. Rank Corr (ρ) is the Spearman correlation of each column's SDG coverage-gap ranks against the canonical MPNet-LR column.

Table 16. Cross-sensitivity robustness of within-SDG coverage-gap rankings across policy source and retrieval strategy.

SDG		Policy source			Retrieval	
	Canon	Curated	SDGi	UNGDC	LR	MLP
SDG 1	7	17	4	11	6	6
SDG 2	15	8	12	12	12	13
SDG 3	1	3	1	2	2	2
SDG 4	4	4	3	6	7	14
SDG 5	8	11	7	17	8	10
SDG 6	9	15	9	16	9	8
SDG 7	16	7	17	8	17	17
SDG 8	11	9	8	9	15	15
SDG 9	2	1	2	4	1	1
SDG 10	14	16	13	15	14	9
SDG 11	17	5	15	7	10	7
SDG 12	12	14	10	10	16	16
SDG 13	5	6	16	5	5	5
SDG 14	10	13	14	13	11	11
SDG 15	13	10	11	14	13	12
SDG 16	6	12	6	1	4	4
SDG 17	3	2	5	3	3	3
Rank Corr (ρ)	1.00	0.40	0.77	0.54	0.87	0.66

Notes: Each cell reports the within-SDG coverage-gap rank (1 = largest gap, 17 = smallest gap). The Canon column is the canonical MPNet—LR ranking. Coverage gap = $|\text{research proportion} - \text{policy proportion}|$ per SDG, using document-weighted policy proportions (Assumption A19). Policy-source columns compare the keyword-retrieved research profile against each policy-source family. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. The Segment-cap axis is omitted: coverage gap is segment-cap-independent. Rank Corr (ρ) is the Spearman correlation of each column's SDG coverage-gap ranks against the canonical column.

Table 17. Coverage gap values across embedding architectures, assignment methods, and retrieval strategies.

SDG	Coverage gap (%)							
	MPNet			MiniLM		SciBERT		Concept
	LR	MLP	ZS	LR	MLP	LR	MLP	
1	4.1	4.5	3.8	4.1	4.1	5.1	5.6	3.9
2	1.9	1.5	0.7	2.1	2.3	2.6	2.9	2.5
3	21.0	16.8	17.2	22.0	22.4	22.3	23.2	21.0
4	10.8	8.9	11.7	8.9	8.8	12.8	12.3	3.5
5	3.1	2.2	2.6	1.7	2.4	3.9	3.4	3.3
6	2.7	2.6	1.1	2.3	2.3	2.5	2.4	2.9
7	0.2	0.4	4.0	1.0	0.8	0.6	0.1	0.4
8	2.1	1.1	1.9	2.3	2.5	3.7	2.4	2.0
9	15.8	15.0	9.4	12.5	15.1	17.6	11.6	21.6
10	2.0	2.2	1.3	1.1	1.6	1.6	1.5	2.1
11	0.2	0.5	0.3	1.6	1.6	1.8	1.6	2.7
12	2.1	1.1	1.0	1.6	1.6	2.5	1.0	0.9
13	7.6	5.7	8.5	5.1	5.9	5.9	8.6	7.1
14	2.2	1.9	0.2	2.2	2.2	2.8	2.4	2.5
15	2.0	1.9	0.1	0.1	1.0	1.3	0.4	2.1
16	6.1	3.9	1.7	11.2	11.1	9.1	4.8	8.5
17	11.4	12.1	20.2	8.9	8.5	9.1	13.2	11.5

Notes: Coverage gap = |research proportion − policy proportion| per SDG in percentage points, using document-weighted policy proportions (Assumption A19) for all methods.

Table 18. (continued) Semantic gap values across embedding architectures, assignment methods, and retrieval strategies.

SDG	Semantic gap (cosine)							
	MPNet			MiniLM		SciBERT		Concept
	LR	MLP	ZS	LR	MLP	LR	MLP	
1	0.415	0.413	0.407	0.222	0.235	0.032	0.061	0.827
2	0.299	0.330	0.345	0.297	0.306	0.037	0.041	0.428
3	0.494	0.480	0.436	0.495	0.507	0.056	0.056	0.663
4	0.267	0.267	0.265	0.291	0.298	0.044	0.043	0.402
5	0.278	0.380	0.246	0.470	0.360	0.041	0.040	0.575
6	0.292	0.292	0.428	0.298	0.294	0.045	0.046	0.384
7	0.428	0.429	0.525	0.430	0.414	0.052	0.053	0.514
8	0.387	0.507	0.278	0.336	0.317	0.033	0.045	0.625
9	0.314	0.327	0.243	0.350	0.384	0.045	0.041	0.421
10	0.271	0.317	0.305	0.485	0.434	0.047	0.049	0.458
11	0.449	0.510	0.477	0.397	0.371	0.052	0.057	0.693
12	0.296	0.438	0.369	0.391	0.411	0.039	0.052	0.400
13	0.480	0.525	0.423	0.541	0.526	0.053	0.051	0.598
14	0.358	0.431	0.452	0.330	0.334	0.044	0.046	0.504
15	0.239	0.231	0.325	0.345	0.307	0.044	0.045	0.346
16	0.329	0.332	0.294	0.322	0.322	0.040	0.037	0.541
17	0.177	0.267	0.398	0.241	0.355	0.023	0.030	0.312

Notes: Semantic gap $= 1 - \cos(\mathbf{r}_{\mathrm{SDG}}, \mathbf{p}_{\mathrm{SDG}})$ where $\mathbf{r}_{\mathrm{SDG}}$ and $\mathbf{p}_{\mathrm{SDG}}$ are the research and policy centroids; policy centroids use a 50-segment-per-document cap. All values are cosine units. This is a *values* table reporting the adjusted (register-removed) semantic gap only; it carries no raw/adjusted panel split, which is shown in the rank-robustness tables (Section I.2).

Table 19. Assignment agreement between the supervised LR classifier and the zero-shot nearest-centroid rule, by SDG and overall, under the canonical MPNet encoder and keyword retrieval.

	Research (papers)			Policy (segments)		Semantic-gap rank		
SDG	n_{LR}	Agree %	Res MLP %	n_{LR}	Agree %	LR	ZS	$\lvert\Delta\rvert$
SDG 1 (No Poverty)	13,459	30.9	14.9	1,700	70.8	5	7	2
SDG 2 (Zero Hunger)	78,591	77.6	59.8	1,723	94.1	10	10	0
SDG 3 (Good Health)	859,355	75.0	79.8	2,934	87.7	1	4	3
SDG 4 (Quality Education)	470,002	82.4	82.6	2,055	91.3	15	15	0
SDG 5 (Gender Equality)	64,940	56.3	37.0	2,026	88.5	13	16	3
SDG 6 (Clean Water)	35,137	85.3	75.2	1,206	94.6	12	5	7
SDG 7 (Clean Energy)	119,467	88.3	81.9	1,808	84.1	4	1	3
SDG 8 (Decent Work)	65,500	44.4	20.7	2,016	71.3	6	14	8
SDG 9 (Industry & Infra.)	577,833	50.8	45.1	4,066	81.0	9	17	8
SDG 10 (Reduced Inequalities)	30,323	23.6	12.8	1,123	56.9	14	12	2
SDG 11 (Sustainable Cities)	146,508	77.2	64.0	1,658	88.3	3	2	1
SDG 12 (Responsible Cons.)	46,199	60.9	38.5	1,187	91.8	11	9	2
SDG 13 (Climate Action)	121,557	40.5	28.4	3,399	84.5	2	6	4
SDG 14 (Life Below Water)	51,458	89.6	69.7	1,634	92.3	7	3	4
SDG 15 (Life on Land)	78,102	83.5	70.3	1,828	84.1	16	11	5
SDG 16 (Peace & Justice)	337,317	55.6	46.1	5,725	65.0	8	13	5
SDG 17 (Partnerships)	9,396	26.1	11.7	4,509	84.1	17	8	9
Overall	3,105,144	67.3	59.5	40,597	81.5			

Notes: Agree % is the share of LR-assigned texts that the zero-shot rule also assigns to the same SDG. Semantic-gap ranks are 1 = largest gap; $|\Delta|$ is the absolute rank difference between the two methods. Assignment agreement is computed over all texts (research: each paper; policy: each segment), without the 50-segment-per-document cap, which applies only to centroid building. The zero-shot rule assigns each text to the SDG with the highest cosine similarity to the train-split reference centroids ($\mathbf{s}_k$); the LR rule uses the trained classifier's argmax. MLP agreement against zero-shot is computed for the policy corpus only, because MLP research per-paper scores are not persisted (the research LR-vs-MLP gap-rank correlation is reported in Section I.3).

Table 20. Pearson r on the raw coverage and semantic-gap magnitudes.

	Raw gap	Adj. gap	Register
H1a. Coverage gap ↔ Semantic gap			
MPNet LR	+0.102	+0.694**	-0.382
MPNet MLP	-0.069	+0.441†	-0.404
MPNet ZS	-0.171	+0.230	-0.333
MiniLM LR	+0.068	+0.474†	-0.226
MiniLM MLP	+0.333	+0.611**	-0.093
SciBERT LR	+0.129	+0.152	-0.080
SciBERT MLP	-0.108	-0.189	+0.138
Concept LR	+0.030	+0.445†	-0.357
Concept MLP	-0.092	+0.196	-0.310
H1b. Policy–research dominance ↔ Semantic gap			
MPNet LR	+0.274	+0.196	+0.115
MPNet MLP	+0.050	+0.090	-0.014
MPNet ZS	-0.152	-0.227	+0.033
MiniLM LR	+0.293	+0.119	+0.240
MiniLM MLP	+0.353	+0.150	+0.260
SciBERT LR	+0.460†	-0.156	+0.420†
SciBERT MLP	+0.194	-0.358	+0.484*
Concept LR	+0.117	+0.120	+0.021
Concept MLP	-0.060	+0.046	-0.119
H1c. Research coverage ↔ Semantic gap			
MPNet LR	+0.288	+0.464†	-0.055
MPNet MLP	+0.032	+0.308	-0.197
MPNet ZS	-0.191	-0.117	-0.087
MiniLM LR	+0.302	+0.355	+0.101
MiniLM MLP	+0.429†	+0.438†	+0.132
SciBERT LR	+0.419†	+0.208	+0.029
SciBERT MLP	+0.029	-0.234	+0.262
Concept LR	+0.134	+0.262	-0.085
Concept MLP	-0.074	+0.113	-0.205
H1d. Policy coverage ↔ Semantic gap			
MPNet LR	-0.018	+0.501*	-0.359
MPNet MLP	-0.045	+0.418†	-0.361
MPNet ZS	-0.052	+0.258	-0.244

Table 21. Per-SDG policy source-family sensitivity: coverage share and semantic gap.

SDG	Full Corpus		Curated AI/SDG		SDGi VNR/VLR		UNGDC	
	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)
1	4.5 (288)	0.42 (1,689)	1.1 (1)	0.44 (357)	5.9 (250)	0.42 (1,191)	1.8 (37)	0.56 (141)
2	4.3 (274)	0.30 (1,508)	0.0 (0)	0.35 (240)	5.8 (246)	0.29 (1,196)	1.4 (28)	0.45 (72)
3	4.7 (298)	0.49 (2,408)	6.4 (6)	0.36 (643)	6.7 (284)	0.55 (1,707)	0.4 (8)	0.70 (58)
4	4.8 (305)	0.27 (2,055)	0.0 (0)	0.21 (274)	7.0 (296)	0.29 (1,718)	0.4 (9)	0.40 (63)
5	5.1 (324)	0.28 (2,005)	0.0 (0)	0.28 (272)	6.6 (278)	0.28 (1,531)	2.2 (46)	0.41 (202)
6	3.7 (238)	0.29 (1,206)	0.0 (0)	0.35 (137)	5.3 (226)	0.29 (1,038)	0.6 (12)	0.42 (31)
7	4.3 (273)	0.43 (1,578)	1.1 (1)	0.51 (358)	6.0 (253)	0.41 (1,147)	0.9 (19)	0.59 (73)
8	4.3 (276)	0.39 (1,886)	0.0 (0)	0.40 (309)	6.5 (275)	0.40 (1,558)	0.0 (1)	0.54 (19)
9	4.3 (274)	0.31 (2,986)	50.0 (47)	0.27 (1,838)	5.1 (215)	0.54 (1,116)	0.6 (12)	0.46 (32)
10	2.9 (182)	0.27 (1,085)	0.0 (0)	0.33 (258)	4.2 (176)	0.28 (774)	0.3 (6)	0.40 (53)
11	5.3 (338)	0.45 (1,658)	0.0 (0)	0.43 (175)	8.0 (336)	0.46 (1,474)	0.1 (2)	0.57 (9)
12	3.5 (226)	0.30 (1,132)	0.0 (0)	0.38 (178)	5.3 (225)	0.30 (947)	0.0 (1)	0.49 (7)
13	11.3 (718)	0.48 (3,084)	7.4 (7)	0.44 (613)	6.0 (252)	0.44 (1,268)	22.4 (459)	0.58 (1,203)
14	3.8 (241)	0.36 (1,293)	0.0 (0)	0.37 (216)	4.5 (189)	0.35 (856)	2.5 (52)	0.49 (221)
15	4.2 (266)	0.24 (1,484)	0.0 (0)	0.29 (299)	5.7 (239)	0.24 (1,108)	1.3 (27)	0.41 (77)
16	17.3 (1,100)	0.33 (4,890)	12.8 (12)	0.20 (1,092)	5.9 (250)	0.38 (1,395)	40.9 (838)	0.51 (2,403)
17	11.7 (746)	0.18 (3,645)	21.3 (20)	0.18 (859)	5.6 (235)	0.16 (1,322)	24.0 (491)	0.26 (1,464)
ρ	1.00	1.00	0.36	0.73	0.61	0.91	0.45	0.97

Notes: For each policy source family, `pol.% (n)` is the document-weighted policy share (%) with the document count in parentheses; `sem. (n)` is the within-SDG research–policy semantic gap with the capped segment count in parentheses. Read across a row to see whether the gap is stable or driven by one family.

Table 22. Per-family interaction test: coverage predictors vs the within-SDG semantic gap.

Family	n	Research share (ρ, p)	Coverage gap (ρ, p)	Research share excl. SDG 4 (ρ, p)
Full corpus	17	0.45 (0.071)	-0.01 (0.963)	0.61 (0.013)
Curated AI/SDG	17	-0.06 (0.830)	-0.22 (0.390)	0.05 (0.846)
SDGi VNR/VLR	17	0.60 (0.012)	0.04 (0.874)	0.70 (0.002)
UNGDC speeches	15	0.42 (0.114)	0.35 (0.196)	0.61 (0.021)

Notes: Spearman ρ (with two-sided p) between each coverage predictor and the within-SDG semantic gap across the reliable SDGs (n). Research share is the research-corpus proportion per SDG (family-independent); the coverage gap is |research − policy| using the family's document-weighted policy share. The curated AI/SDG family is the symmetric AI/SDG-vs-AI/SDG control matching the research corpus's AI ∩ SDG scope.

J Pooled Regression: Coverage Predictors and the Semantic Gap

This appendix pools all 24 encoder–retrieval–method–cap configurations into a single panel ($N = 408$) and regresses the within-SDG semantic gap on coverage predictors using OLS with cluster-robust standard errors by SDG (17 clusters). The dependent variable is the rank of the semantic gap within each configuration (1–17 per config, ties averaged); predictors are expressed in percentage points (0–100). Table 23 reports the full 23-specification grid across core regressions (Panel A), robustness checks (Panel B), interaction terms (Panel C), and alternative functional forms (Panel D).

The adjusted (topic) gap is positively associated with the coverage gap ($b = +0.18$, $p = 0.069$; Panel A, column 1); the coefficient is marginally significant at conventional thresholds but is confirmed by the supervised-only subsample ($p = 0.040$), the MPNet-only subsample ($p = 0.003$), and the raw-DV specification in cosine units ($p = 0.007$). The raw gap and the register gap show no detectable association with any coverage predictor, consistent with the cancellation pattern reported in Section 4.3. Policy coverage (the share of policy documents assigned to a given SDG) is a stronger predictor of the adjusted gap ($b = +0.41$, $p = 0.003$), capturing how concentrated policy attention is on each goal. The coverage-gap-by-encoder interaction (Panel C) reveals that the effect is strongest for MPNet and reverses for SciBERT, suggesting that domain-specialised pretraining compresses the topic gap regardless of coverage structure.

Table 23. Pooled OLS: coverage predictors and the within-SDG semantic gap rank across 24 configurations.

	Core									Robustness						Interactions				Functional form			
	(1) Adj. gap covgap	(2) Adj. gap dominance	(3) Raw gap covgap	(4) Raw gap dominance	(5) Reg. gap covgap	(6) Reg. gap dominance	(7) Adj. gap rescov	(8) Raw gap rescov	(9) Reg. gap rescov	(10) No classifier ind.	(11) + SDG FE	(12) Supervised only	(13) Keyword only	(14) MPNet only	(15) No clf + SDG FE	(16) + covgap×enc	(17) + covgap×ret	(18) + covgap×mth	(19) + all interactions	(20) Raw DV (not ranked)	(21) Log DV	(22) WLS	(23) Rank + bootstrap
Coverage gap ($\lvert$research% − policy%$\rvert$)	+0.2†	–	+0.1	–	-0.0	–	–	–	–	+0.2†	-0.0	+0.2*	+0.2	+0.3**	-0.0	+0.3**	+0.2	+0.2**	+0.4***	+0.0**	+0.0**	+0.4***	+0.2†
	(0.1)		(0.2)		(0.1)					(0.1)	(0.2)	(0.1)	(0.1)	(0.1)	(0.2)	(0.1)	(0.1)	(0.1)	(0.1)	(0.0)	(0.0)	(0.1)	(0.1)
Policy coverage	+0.4**	+0.5**	-0.1	+0.1	-0.3†	-0.3	+0.5**	-0.1	-0.3†	+0.4**	+1.2†	+0.4**	+0.4**	+0.4*	+1.2†	+0.4**	+0.4**	+0.4**	+0.4**	+0.0*	+0.0*	+0.5***	+0.4**
	(0.1)	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)	(0.1)	(0.2)	(0.2)	(0.1)	(0.7)	(0.1)	(0.1)	(0.1)	(0.7)	(0.1)	(0.1)	(0.1)	(0.1)	(0.0)	(0.0)	(0.1)	(0.1)
MiniLM (vs. MPNet)	+0.1	-0.0	+0.0	-0.0	-0.0	+0.0	-0.0	-0.0	+0.0	+0.1	-0.0	+0.1	+0.1	–	-0.0	+0.4	+0.1	+0.1	+0.5	-0.0**	-0.0**	+0.2	+0.1
	(0.8)	(0.8)	(1.2)	(1.1)	(0.9)	(0.9)	(0.8)	(1.1)	(0.9)	(0.8)	(0.8)	(0.9)	(0.8)		(0.8)	(1.1)	(0.8)	(0.8)	(1.1)	(0.0)	(0.0)	(0.9)	(0.8)
SciBERT (vs. MPNet)	-0.1	-0.0	-0.0	-0.0	+0.0	+0.0	-0.0	-0.0	+0.0	-0.1	+0.0	-0.1	-0.1	–	+0.0	+2.6*	-0.1	-0.1	+2.8*	-0.1***	-0.1***	-3.4	-0.1
	(1.1)	(1.1)	(1.0)	(1.0)	(1.3)	(1.3)	(1.1)	(1.0)	(1.3)	(1.1)	(1.1)	(1.3)	(1.1)		(1.1)	(1.2)	(1.1)	(1.1)	(1.2)	(0.0)	(0.0)	(2.1)	(1.1)
Concept (vs. keyword)	-0.0	+0.0	-0.0	-0.0	+0.0	-0.0	+0.0	+0.0	-0.0	-0.0	+0.0	-0.0	–	-0.1	+0.0	-0.1	-0.6	-0.0	+0.8	+0.1***	+0.1***	-1.0	-0.0
	(0.7)	(0.7)	(0.8)	(0.8)	(1.0)	(1.0)	(0.7)	(0.8)	(1.0)	(0.6)	(0.7)	(0.7)		(0.8)	(0.7)	(0.7)	(1.0)	(0.7)	(0.8)	(0.0)	(0.0)	(1.2)	(0.7)
Cap 20 (vs. 50)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	+0.0	-0.0	+0.0	-0.0	+0.0	-0.0	+0.0	-0.0	-0.0	-0.0	+0.0*	+0.0*	-0.1	-0.0
	(0.1)	(0.1)	(0.1)	(0.1)	(0.2)	(0.2)	(0.1)	(0.1)	(0.2)	(0.2)	(0.1)	(0.1)	(0.1)	(0.1)	(0.2)	(0.1)	(0.1)	(0.1)	(0.1)	(0.0)	(0.0)	(0.3)	(0.1)
No cap (vs. 50)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	+0.0	+0.0	-0.0	+0.0	-0.0	+0.0	-0.0	-0.0	+0.0	-0.0	+0.0	+0.0	+0.0	-0.2	-0.0
	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)	(0.3)	(0.2)	(0.2)	(0.3)	(0.2)	(0.3)	(0.2)	(0.2)	(0.2)	(0.2)	(0.0)	(0.0)	(0.4)	(0.2)
MLP (vs. LR)	-0.0	-0.0	-0.0	+0.0	-0.0	-0.0	+0.0	+0.0	+0.0	–	-0.0	+0.0	-0.0	+0.0	–	-0.0	-0.0	+0.6	+0.5	+0.0*	+0.0*	-0.8	-0.0
	(0.4)	(0.4)	(0.5)	(0.5)	(0.3)	(0.3)	(0.4)	(0.5)	(0.3)		(0.4)	(0.4)	(0.4)	(0.6)		(0.4)	(0.4)	(0.5)	(0.5)	(0.0)	(0.0)	(0.9)	(0.4)
ZS (vs. LR)	+0.0	+0.0	+0.0	-0.0	+0.0	-0.0	+0.0	+0.0	-0.0	–	-0.0	–	+0.0	+0.0	–	-0.0	-0.0	+0.8	+0.7	+0.0	+0.0	-0.8	+0.0
	(0.7)	(0.7)	(1.3)	(1.3)	(0.8)	(0.8)	(0.7)	(1.3)	(0.8)		(0.7)		(0.7)	(1.0)		(0.7)	(0.7)	(0.9)	(0.9)	(0.0)	(0.0)	(1.3)	(0.7)
Domination (research% − policy%)	–	+0.1	–	+0.1	–	+0.0	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
		(0.1)		(0.1)		(0.1)																	
Research coverage (%)	–	–	–	–	–	–	+0.1	+0.1	+0.0	–	–	–	–	–	–	–	–	–	–	–	–	–	–
							(0.1)	(0.1)	(0.1)														
sdg1	–	–	–	–	–	–	–	–	–	–	+7.5†	–	–	–	+7.5†	–	–	–	–	–	–	–	–
											(3.9)				(3.9)								
sdg2	–	–	–	–	–	–	–	–	–	–	-2.8	–	–	–	-2.8	–	–	–	–	–	–	–	–
											(4.3)				(4.3)								
sdg3	–	–	–	–	–	–	–	–	–	–	+7.3	–	–	–	+7.3	–	–	–	–	–	–	–	–
											(4.5)				(4.5)								
sdg4	–	–	–	–	–	–	–	–	–	–	+1.5	–	–	–	+1.5	–	–	–	–	–	–	–	–
											(3.6)				(3.6)								
sdg5	–	–	–	–	–	–	–	–	–	–	+4.8	–	–	–	+4.8	–	–	–	–	–	–	–	–
											(3.7)				(3.7)								
sdg6	–	–	–	–	–	–	–	–	–	–	-1.0	–	–	–	-1.0	–	–	–	–	–	–	–	–
											(4.6)				(4.6)								
sdg7	–	–	–	–	–	–	–	–	–	–	+1.6	–	–	–	+1.6	–	–	–	–	–	–	–	–
											(4.4)				(4.4)								
sdg8	–	–	–	–	–	–	–	–	–	–	+2.0	–	–	–	+2.0	–	–	–	–	–	–	–	–
											(4.2)				(4.2)								
sdg9	–	–	–	–	–	–	–	–	–	–	+1.5	–	–	–	+1.5	–	–	–	–	–	–	–	–
											(4.1)				(4.1)								
sdg10	–	–	–	–	–	–	–	–	–	–	+10.1†	–	–	–	+10.1†	–	–	–	–	–	–	–	–
											(5.2)				(5.2)								
sdg11	–	–	–	–	–	–	–	–	–	–	+1.9	–	–	–	+1.9	–	–	–	–	–	–	–	–
											(3.6)				(3.6)								
sdg12	–	–	–	–	–	–	–	–	–	–	+3.5	–	–	–	+3.5	–	–	–	–	–	–	–	–
											(4.7)				(4.7)								
sdg13	–	–	–	–	–	–	–	–	–	–	-1.6†	–	–	–	-1.6†	–	–	–	–	–	–	–	–
											(0.8)				(0.8)								
sdg14	–	–	–	–	–	–	–	–	–	–	+0.8	–	–	–	+0.8	–	–	–	–	–	–	–	–
											(4.6)				(4.6)								
sdg15	–	–	–	–	–	–	–	–	–	–	-2.8	–	–	–	-2.8	–	–	–	–	–	–	–	–
											(4.4)				(4.4)								
sdg16	–	–	–	–	–	–	–	–	–	–	-9.4	–	–	–	-9.4	–	–	–	–	–	–	–	–
											(5.7)				(5.7)								
covgap×i_{minilm}	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	-0.0	–	–	-0.1	–	–	–	–
																(0.1)			(0.1)				
covgap×i_{scibert}	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	-0.5***	–	–	-0.5***	–	–	–	–
																(0.1)			(0.1)				
covgap×i_{concept}	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	+0.1	–	-0.1†	–	–	–	–
																	(0.1)		(0.1)				
covgap×i_{mlp}	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	-0.1†	-0.1†	–	–	–	–
																		(0.0)	(0.0)				
covgap×i_{zs}	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	-0.1	-0.1	–	–	–	–
																		(0.1)	(0.1)				
N	408	408	408	408	408	408	408	408	408	408	408	357	357	170	408	408	408	408	408	408	408	408	408
R^2	0.195	0.162	0.014	0.049	0.073	0.074	0.162	0.049	0.074	0.195	0.617	0.198	0.196	0.331	0.617	0.255	0.197	0.200	0.260	0.684	0.701	0.521	0.195
Adj. R^2	0.177	0.143	-0.008	0.028	0.052	0.053	0.143	0.028	0.053	0.181	0.592	0.179	0.177	0.302	0.594	0.235	0.177	0.178	0.233	0.677	0.694	0.510	0.177

K Declaration of AI Use

In accordance with the University's guidance on generative AI ("Use of GenAI is permitted to support assignments"), this appendix declares the use of AI tools throughout the dissertation. All use was conducted with human oversight and control; AI-generated outputs were reviewed, modified, and independently verified before inclusion. No AI-generated text was submitted without such verification. This declaration follows the order of permitted activities specified in the assessment guidelines. AI-assisted activities not explicitly listed (e.g., code debugging, plotting, LaTeX troubleshooting, literature summarisation) fall under the same oversight and verification standards described in the relevant rows above.

Table 24. Declaration of AI use by permitted activity.

Activity	AI tool(s)	How output was used
Researching and refining ideas	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	ChatGPT was used iteratively to develop ideas that aligned with the author's intent. Coding agents independently checked data and code availability to confirm feasibility. All final decisions and conclusions are the author's own. Final research design was formally presented to and approved by supervisor.
Information retrieval / background research	Gemini, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	AI tools read the manuscript for context and recommended papers to investigate further. All cited sources were independently verified against the original publications.
Drafting an outline / organising thoughts	MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Generated code scaffolding and assisted with the full analysis pipeline (embedding, scoring, validation, robustness, appendix outputs). The centroid-assignment step was verified against hand-labeled samples, and the scoring and coverage-gap functions were unit-tested. Analytical decisions were directed by the author.
Refining research questions	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Overlaps with row 1. ChatGPT sharpened framing based on reviewed literature; coding agents confirmed questions were answerable with available data and models. The author retained full control over the final formulation. RQs presented to and approved by supervisor.
Checking spelling and grammar	ChatGPT, Gemini, Claude (web/mobile), MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Coding agents provided inline stylistic and phrasing suggestions during drafting. Web and mobile versions of ChatGPT, Gemini, and Claude were used for editorial review and proofreading. All suggestions were accepted or rejected per standard editorial judgment.